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Nonparametric predictive regression for stock return prediction

Jiti Gao^b, Tingting Cheng^a, Oliver Linton^c, and Yayi Yan^d

^aSchool of Finance, Nankai University, Tianjin, China; ^bDepartment of Econometrics and Business Statistics, Monash University, Melbourne, Victoria, Australia; ^cFaculty of Economics, University of Cambridge, Cambridge, UK; ^dSchool of Statistics and Data Science, Shanghai University of Finance and Economics, Shanghai, China

ABSTRACT

We propose a multi-step nonparametric predictive regression model, which allows for general locally stationary predictors and time-varying/nonlinear return predictability. We propose a kernel estimation method and establish the large sample properties in both short and long horizons. We apply our method to stock return prediction using a number of standard predictors such as dividend yield. The empirical results show that our proposed model can outperform the historical mean benchmark, linear predictive regression model, and several machine learning methods for some cases in terms of out-of-sample forecasting performance. We also compare our method with the historical mean benchmark using an economic metric. In particular, we show how our methods could be used to deliver a trading strategy that beats the buy-and-hold strategy over our sample period.

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1. Introduction

A fundamental issue in finance is whether future stock returns are predictable using publicly available information. The seminal studies of Keim and Stambaugh (1986), Fama and French (1988), and Campbell and Shiller (1988) empirically demonstrate that variables such as dividend yield, book-to-market ratio, or term spread have significant predictive ability for future stock returns using data up to the early 1980's. A systematic discussion on the performance of linear predictive models is given by Welch and Goyal (2008), in which however they find that these linear models predict stock returns poorly both in-sample (IS) and out-of-sample (OOS), and that these models seem unstable with respect to OOS forecasting experiments.

Regarding the IS evidence of return predictability, perhaps the most important issue is the uncertainty about the persistence of the selected predictors. For example, in the case of a unit-root predictor but wrongly assuming stationarity, the IS test for predictability tends to over-reject the null hypothesis and makes the “significance” results spurious (Stambaugh, 1999; Campbell and Yogo, 2006). In addition, while the conventional unit-root tests suggest the commonly used predictors contain unit roots (Phillips, 2015), the unit-root assumption may not make much economic sense and a large body of empirical evidence supports the no bubble condition for asset prices (e.g., Giglio, Maggiori, and Stroebl, 2016). To address this issue, Yu and Yan (2023) argue that the high persistence of the commonly used predictors is due to their time-varying levels and then propose a locally stationary autoregressive process to model the dynamics of predictors. Empirically, Fama and French (2001) document the phenomenon of so-called “disappearing dividends”, i.e., the dividend-price ratio (the most commonly used predictor in the literature) is gradually declining since the 1970s, which supports the assumption of local stationarity.

CONTACT Jiti Gao  jiti.gao@gmail.com  Department of Econometrics and Business Statistics, Monash University, Melbourne, VIC, Australia.

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With respect to the OOS forecasting performance of the predictors, it is shown in the recent literature that return predictability varies over time (Dangl and Halling, 2012; Cujean and Hasler, 2017; Farmer, Schmidt, and Timmermann, 2023). This stylized fact also motivates the use of time-varying coefficient models. Specifically, by using a nonparametric time-varying model, Farmer, Schmidt, and Timmermann (2023) find that stock returns are predictable in some local time but are interspersed with long periods of no return predictability. In addition, they find that this phenomenon is consistent with a sticky expectations model in which investors slowly update their beliefs about a persistent component in the cash flow process. On the other hand, some other researchers consider nonlinear/nonparametric models and machine learning methods to predict stock returns (e.g., Scholz, Nielsen, and Sperlich, 2015; Scholz, Sperlich, and Nielsen, 2016; Gu, Kelly, and Xiu, 2020; Jacobsen, Jiang, and Zhang, 2020). For example, Scholz, Nielsen, and Sperlich (2015) demonstrate that prediction of excess stock returns can essentially be improved by using flexible non and semi-parametric techniques. Using a single-index predictive regression model, Zhou et al. (2024) document some nonlinear return predictability with respect to the value of the predictors. Gu, Kelly, and Xiu (2020) employ a comprehensive set of machine learning tools to predict monthly individual stock returns using firm specific and common predictors and find that the predictive gains of these machine learning tools is due to allowing nonlinear predictor interactions.

To address the aforementioned issues in a unified framework, this article considers a nonparametric approach that allows for general locally stationary predictors, as well as time-varying and nonlinear predictability. To be specific, we propose a new predictive model: the multi-step nonparametric predictive regression model (NPR), which allows for nonlinear predictive relation of unknown form between stock returns and the selected predictors. We also use rescaled time as one of our covariates, which allows for variation over time in the predictive relationship, a point emphasized by for example Farmer, Schmidt, and Timmermann (2023). In addition, we allow the predictive variables to be general locally stationary processes, which carries robustness with respect to the changing time-series properties of the predictors, rules out the bubbles in asset prices, and captures time variation in the mean and persistence of predictors. We present the theoretical properties of our estimators of the regression functions in both short horizon and long horizon cases, where by long horizon we mean that the horizon grows to infinity with the size of the sample. Many empirical studies consider the long horizon case using a linear predictive model (e.g., Phillips and Lee, 2013; Xu, 2020) and our results support the use of nonparametric methods, thereby allowing for time-varying and nonlinear long horizon predictability.

To evaluate the effectiveness of our proposed NPR model, we investigate its capability of monthly stock return prediction over the period 1965–2018. The empirical results show that our new model can substantially outperform the historical mean benchmark, traditional linear predictive regression model, and several machine learning methods for some cases in terms of OOS forecasting performance. Overall, we have found that (1) the out-of-sample forecasting performance of all considered methods varies significantly over time, indicating time-varying return predictability and (2) the performance of NPR combination forecasts is more stable and robust for different out-of-sample periods and there indeed exist evidence of stock return predictability in some out-of-sample periods, though it is noteworthy that the events of 2009 had a great effect on the forecasting performance of NPR models. The outlook for nonparametric methods looks somewhat more promising than the one that was presented in Diebold and Nason (1990), although we acknowledge that the magnitude of the gain provided by these methods is modest. To quantify the economic benefits of our methodology, we define a trading strategy based on our fitting methods. We show that with appropriate choice of tuning parameter our strategy outperforms the buy and hold method, which corresponds to historical mean predictor.

The rest of this article is organized as follows. In Section 2, we describe our proposed NPR model in detail and establish asymptotic properties for the nonparametric estimator of the predictive functions. Section 3 presents the simulation results. In Section 4, we compare the performance of NPR on the prediction of stock returns with traditional historical mean benchmark and linear predictive model. Section 5 concludes the article. Implementation details of the NPR model, particularly the bandwidth selection for this model, and the proofs of the main results are given in Appendix.

2. NPR model and estimation theory

In this section, we first introduce the NPR model and then establish the corresponding estimation theory and asymptotic properties.

2.1. The setup

Consider a nonparametric predictive regression model of the form

$$y_{t+j} = g_j(\tau_t, x_t) + e_{t+j}, \quad t = 1, 2, \dots, n, \quad j = 1, 2, \dots, J, \quad (1)$$

where $\tau_t = t/n$,¹ $x_t = (x_t^1, \dots, x_t^d)^\top$ is a vector of locally stationary predictors, $\{g_j(\cdot)\}$ are unknown functions of τ_t and x_t , and e_{t+j} follows an α -mixing error process. This model allows for time-varying predictability, and nonlinearity between stock returns and the predictors x_t . Typically, y_t is log stock returns, but we may also be interested in predicting exchange rate or volatility.

We are also interested in predicting long horizon returns $\sum_{j=1}^J y_{t+j}$ using the covariates available up to and including time t . It follows from our specification that

$$y_{t:t+J} = \sum_{j=1}^J y_{t+j} = \sum_{j=1}^J g_j(\tau_t, x_t) + \sum_{j=1}^J e_{t+j} = g(\tau_t, x_t) + e_{t:t+J}, \quad (2)$$

where $g(\tau, x) = \sum_{j=1}^J g_j(\tau, x)$ and $e_{t:t+J} = \sum_{j=1}^J e_{t+j}$ both depend on J . Note that the asymptotic theory established for the estimator of $g(\tau, x)$ in [Theorem 2.2](#) below covers the case where $J \rightarrow \infty$ as $n \rightarrow \infty$. Note also however that $\text{cov}(e_{t:t+J}, e_{s:s+J}) \neq 0$ when $|t - s| < J$, which must be allowed for in the distribution theory.

2.2. Estimation procedure

For each fixed j and a given point (τ, x) , we use the local constant kernel method² to estimate $g_j(\tau, x)$ by

$$\widehat{g}_j(\tau, x) = \sum_{t=1}^n W_{nt}(\tau, x; h_j) y_{t+j} \quad \text{with} \quad W_{nt}(\tau, x; h_j) = \frac{K\left(\frac{\tau_t - \tau}{h_j}\right) \prod_{i=1}^d K\left(\frac{x_t^i - x^i}{h_j}\right)}{\sum_{s=1}^n K\left(\frac{\tau_s - \tau}{h_j}\right) \prod_{i=1}^d K\left(\frac{x_s^i - x^i}{h_j}\right)}, \quad (3)$$

where $x = (x^1, \dots, x^d)^\top$ for any vector $x \in R^d$, $K(\cdot)$ is a kernel function and h_j is a bandwidth parameter. For convenience, in this article, we work with a product kernel and assume that the bandwidth h_j is the same for τ and x^i ($i = 1, 2, \dots, d$), but the results can be easily extended to the case involving nonproduct kernels and different bandwidths. We then define our estimator of $g(\tau, x)$ to be the sum of the one dimensional estimators

$$\widehat{g}(\tau, x) = \sum_{j=1}^J \widehat{g}_j(\tau, x). \quad (4)$$

By summing up $\widehat{g}_j(\tau, x)$ to obtain $\widehat{g}(\tau, x)$, we do not need to impose conditions on the accumulated (excess) returns when establishing the theory below. Note that if the bandwidth h_j were constant across j , then $g(\tau, x)$ could be directly estimated from regression (2) using the local constant kernel method,

¹Robinson (1989) demonstrated that this “scaled time” requirement is necessary for the consistency of the nonparametric kernel smoothing, which ensures that by increasing sample size, the number of observations in each shrinking neighborhood increases.

²Following Farmer, Schmidt, and Timmermann (2023), we use local constant estimator here for simplicity of exposition. Of course local linear estimation is another natural choice. The theory developed in this article can also be extended to local linear estimator.

i.e.,

$$\widehat{g}(\tau, x) = \sum_{t=1}^n W_{nt}(\tau, x; h_1) \left(\sum_{j=1}^J y_{t+j} \right).$$

However, as J increases, the convergence rate of $\widehat{g}(\tau, x)$ becomes slower, and a larger h_j is required (still converging to zero). As the bandwidth varies with j , we should use the general formula given by (4).

2.3. Assumptions

A formal definition of locally stationary process is given in the below (cf., Vogt, 2012).

Definition of locally stationary processes: Process $\{x_t\}$ is said to be locally stationary if for each scaled time point $\tau \in [0, 1]$ there exists an associated process $\{x_t(\tau)\}$ satisfying

- (i) $\{x_t(\tau)\}$ is strictly stationary with density $f_{x_t(\tau)}(x)$;
- (ii) it holds that

$$\|x_t - x_t(\tau)\| \leq \left(\left| \frac{t}{n} - \tau \right| + \frac{1}{n} \right) U_{nt}(\tau) \text{ a.s.}, \quad (5)$$

where $U_{nt}(\tau)$ is a process of positive variables such that $\mathbb{E}[U_{nt}^\omega(\tau)] < C$ for some $\omega > 0$ and $C > 0$ independent of τ, t and n , and $\|\cdot\|$ denotes an arbitrary norm on $\mathbb{R}^d$.

It follows from the definition that a stationary process (considered in Chen and Hong, 2010; Farmer, Schmidt, and Timmermann, 2023) is also locally stationary. Under the local stationarity assumption, the predictors (such as the dividend-price ratio) are allowed to be globally nonstationary due to their time-dependent unconditional mean and variance, indicating the presence of a persistent but deterministic time trend. This condition also ensures that the predictors do not contain a unit-root component, thereby ruling out the existence of bubbles in asset prices, which is economically meaningful. Instead, a traditional stationary process is not able to modelling time-varying level and persistence of predictors. Moreover, the framework of local stationarity is also consistent with the finance literature (e.g., Lettau and Van Nieuwerburgh, 2008; Favero, Gozluklu, and Tamoni, 2011), which attributes the highly persistent movements in the commonly used predictors to its time-varying levels.

Before establishing the asymptotic theory, we first introduce some necessary assumptions.

Assumption 2.1. (i) The process $\{x_t\}$ is locally stationary according to the definition above. (ii) It holds that $\max_{j \geq 1} \mathbb{E}|e_{t+j}|^s \leq C$ for some $s \geq 2$ and $C < \infty$. (iii) The array $\{x_t, e_{t+1}, \dots, e_{t+J}\}$ is α -mixing with mixing coefficient α satisfying $\alpha(k) \leq Ak^{-\beta}$ for some $A < \infty$ and $\beta > \frac{2s-2}{s-2}$.

Assumption 2.2. (i) $g_j(\tau, x)$ is twice continuously partially differentiable. (ii) The densities $f(\tau, x) := f_{x_t(\tau)}(x)$ of the variables $x_t(\tau)$ are smooth in τ for each time point $\tau \in [0, 1]$. In particular, $f(\tau, x)$ is differentiable with respect to τ for each $x \in \mathbb{R}^d$, and the derivative $\partial f(\tau, x)/\partial \tau$ is continuous. (iii) $f(\tau, x)$ is partially differentiable with respect to x for each $\tau \in [0, 1]$. The derivatives $\partial f(\tau, x)/\partial x^i$ are continuous for $i = 1, \dots, d$.

Assumption 2.3. Let f_{x_t} and $f_{x_t, x_{t+l}}$ be the densities of x_t and (x_t, x_{t+l}) , respectively. For any compact set $S \subseteq \mathbb{R}^d$, there exists a constant $C = C(S)$ such that $\sup_t \sup_{x \in S} f_{x_t}(x) \leq C$ and $\sup_t \sup_{x \in S} \mathbb{E}[|e_{t+j}|^s | x_t = x] f_{x_t}(x) \leq C$ for the same $s \geq 2$ as in Assumption 2.1. Moreover, there exists a natural number $l^* < \infty$ such that for all $l \geq l^*$, $\sup_t \sup_{x, x' \in S} \mathbb{E}[|e_{t+j}| | e_{t+j+l} | | x_t = x, x_{t+l} = x'] f_{x_t, x_{t+l}}(x, x') \leq C$.

Assumption 2.4. (i) The kernel function $K(\cdot)$ is bounded and has compact support, that is, $K(v) = 0$ for all $|v| > C_1$ with some $C_1 < \infty$. Also, the first moment is zero, that is, $\int v K(v) dv = 0$. Furthermore,

K is Lipschitz continuous, that is, $|K(v) - K(v')| \leq L|v - v'|$ for some $L < \infty$ and all $v, v' \in \mathbb{R}$. (ii) Let $h_j = \rho_j h$, where each ρ_j is a positive constant and $\rho_j \rightarrow \infty$ as $j \rightarrow \infty$; $h \rightarrow 0$ as $n \rightarrow \infty$. In addition, $nh_j^{d+1} \rightarrow \infty$ as $n \rightarrow \infty$.

Assumption 2.1 imposes some weak dependence conditions on the predictors and predictive errors, which allow local stationarity and conditional heteroskedasticity of unknown form. Assumption 2.2 imposes smoothness condition on the unknown functions and the density of $x_t(\tau)$. Assumption 2.3 is required to guarantee a certain rate of the convergence rate. Similar treatments have also been adopted for regression models in the literature (e.g., Vogt, 2012). Assumption 2.4 is a standard assumption for kernel function $K(\cdot)$ and bandwidth h_j .

2.4. Asymptotic theory

Let $f(\tau, x) = f_{x_t(\tau)}(x)$ denote the densities of the variables $x_t(\tau)$. Define $\kappa_0 = \int K^2(u)du$, $\kappa_2 = \int u^2 K(u)du$ and

$$R_j(\tau, x) = \frac{\kappa_2}{2} \sum_{i=1}^d \left(2 \frac{\partial g_j(\tau, x)}{\partial x^i} \frac{\partial f(\tau, x)}{\partial x^i} + \frac{\partial^2 g_j(\tau, x)}{\partial x^{i2}} f(\tau, x) \right) / f(\tau, x), \quad (6)$$

$$b_j(\tau, x) = \frac{\kappa_2}{2} \left(2 \frac{\partial g_j(\tau, x)}{\partial \tau} \frac{\partial f(\tau, x)}{\partial \tau} + \frac{\partial^2 g_j(\tau, x)}{\partial \tau^2} f(\tau, x) \right) / f(\tau, x), \quad (7)$$

We then have the following theorems, while their proofs are given in Appendix.

Theorem 2.1. Assume that Assumptions 2.1–2.4 hold with $\beta \geq 4$. Let $n^r h_j^{d+2} \rightarrow \infty$ with $r = \min\{\omega, 1\}$, in which ω is defined in (5). Moreover, suppose that $f(\tau, x) > 0$ and that $\sigma_j^2(x) = \mathbb{E}[e_{t+j}^2 | x_t = x]$ is continuous. Then for each given j and (τ, x) , as $n \rightarrow \infty$,

$$\sqrt{nh_j^{d+1}} (\widehat{g}_j(\tau, x) - g_j(\tau, x) - B_{j,\tau,x}) \rightarrow_D N(0, V_{j,\tau,x}), \quad (8)$$

where $B_{j,\tau,x} = h_j^2 (R_j(\tau, x) + b_j(\tau, x))$ and $V_{j,\tau,x} = \kappa_0^{d+1} \sigma_j^2(x) / f(\tau, x)$.

It can be shown that the bias of $\widehat{g}_j(\tau, x)$ includes a standard component of order $Op(h_j^2)$ and a nonstandard component of order $Op(n^{-r} h_j^{-d})$ (see Appendix). However, given the assumption $n^r h_j^{d+2} \rightarrow \infty$, the estimation bias resulted from the nonstationarity of regressors (i.e., the part of order $Op(n^{-r} h_j^{-d})$) is asymptotically negligible. As a result, we find that the asymptotic properties of $\widehat{g}_j(\tau, x)$ are very similar to those for the standard local constant estimators with strictly stationary regressors (see Page 63–64 in Chapter 2 of Li and Racine, 2007). Note, however, that although we include rescaled time as a covariate, the large sample variance of the nonparametric estimator depends only on the short run variance of the error term, not on its long run variance. This is because the localization by the stochastic covariate effectively shuffles much of the dependence out of the error term.

Let $h_j = \rho_j h$, $B_j(\tau, x; h) = h^2 \sum_{j=1}^J \rho_j^2 (R_j(\tau, x) + b_j(\tau, x))$, $\Sigma_J(x) = \sum_{j=1}^J \rho_j^{-(d+1)} \sigma_j^2(x)$ and $V(\tau, x) = \kappa_0^{d+1} / f(\tau, x)$, in which $h > 0$ is a bandwidth parameter, ρ_j is a sequence of positive numbers, and $h \rightarrow 0$ as $n \rightarrow \infty$ and $\rho_j \rightarrow \infty$ as $j \rightarrow \infty$.³ We then establish an asymptotic property for $\widehat{g}(\tau, x)$ in the following theorem.

³With the increase of J (diverging to infinity at a slow rate), we obtain a slower rate of convergence of $\widehat{g}(\tau, x)$ and require a larger h_j (but still convergence to zero). Since $h_j = \rho_j h$, we then have $\rho_j \rightarrow \infty$ as $j \rightarrow \infty$.

Theorem 2.2. Let Assumptions 2.1–2.4 hold. Suppose further that $\lim_{n \rightarrow \infty} nh^{d+1} \Sigma_J^{-1}(x) = \infty$ and $\lim_{n \rightarrow \infty} nh^{d+1} \Sigma_J^{-1}(x) B_J^2(\tau, x; h) < \infty$ for each given (τ, x) . Then as $n \rightarrow \infty$,

$$\sqrt{nh^{d+1} \Sigma_J^{-1}(x)} (\widehat{g}(\tau, x) - g(\tau, x) - B_J(\tau, x; h)) \rightarrow_D N(0, V(\tau, x)). \quad (9)$$

Theorems 2.1 and 2.2 show that $\widehat{g}_j(\tau, x)$ is a consistent estimator of $g_j(\tau, x)$ and is asymptotically normally distributed. Theorem 2.2 remains valid regardless of whether J is fixed or varying. In the case $J \rightarrow \infty$, the rate of $J \rightarrow \infty$ is linked implicitly in the conditions of $\frac{nh^{d+1}}{\Sigma_J(x)} \rightarrow \infty$ as $(n, J) \rightarrow (\infty, \infty)$ and $\lim_{n \rightarrow \infty} nh^{d+1} \Sigma_J^{-1}(x) B_J^2(\tau, x; h) < \infty$. For example, when $\sigma_j^2(x) = 1$ and $\rho_j = j^\gamma$ for $0 < \gamma < \frac{1}{d+1}$, the condition of $\frac{nh^{d+1}}{\Sigma_J(x)} \rightarrow \infty$ reduces to requiring $\frac{nh^{d+1}}{j^{1-\gamma(d+1)}} \rightarrow \infty$ as $(n, J) \rightarrow (\infty, \infty)$. This can be satisfied when $J = \lceil n^{\delta_1} \rceil$ for a certain choice of $0 < \delta_1 < 1$ such that $n^{1-\delta_1(1-\gamma(d+1))} h^{d+1} \rightarrow \infty$ as $n \rightarrow \infty$. In general, the long horizon estimator may converge at a slower rate than in the fixed horizon case depending on the rate at which the bias term increases with horizon, which is related to the memory of the predictor process. In the special case where g is linear and x is a stationary autoregressive process, an appropriate choice of ρ_j will ensure that the bias is summable and the rate of convergence in Theorem 2.2 is as in Theorem 2.1. In the parametric case, the large sample variance reflects the use of overlapping data and standard errors need to be adjusted (Hansen and Hodrick, 1980; Hodrick, 1992); in the nonparametric case, we have the “whitening by smoothing” phenomenon, which has been commented on by many authors. Our results show that this continues to hold when the degree of overlap increases with sample size. Consequently, standard error construction is straightforward.

Some details for practical implementations (in particular, the choice of bandwidth h_j and the estimation of asymptotic variance) are presented in Appendices A.1 and A.2 for ease of presentation.

3. Simulation study

In this section, we conduct simulations to examine the theoretical findings of Section 2. Specifically, we consider the following data generating processes.

$$y_{t+j} = g_j(\tau_t, x_t) + e_{t+j}, \quad t = 1, \dots, n, \quad j = 1, \dots, J \quad (10)$$

where $\tau_t = t/n$, e_{t+j} is generated from standard normal distribution and x_t is generated from a locally stationary process $x_t = \mu(\tau_t) + \sigma(\tau_t)\epsilon_t$, in which $\mu(\tau) = \sin(2\pi\tau)$ and $\sigma(\tau) = 1 + 0.5 \cos(2\pi\tau)$ and ϵ_t is generated from standard normal distribution. We set $g_j(\tau, x) = \sin(2\pi\tau) + \cos(2\pi x)$.

We consider sample sizes of $n = 100, 200, 400$ and let $R = 500$ be the number of replications. Let $g_j(\tau, x)$ and $\widehat{g}_j^{(r)}(\tau_t, x_t)$ denote the true value and the estimated value of the $g_j(\cdot)$ at the r -th replication, respectively. Under each sample size, we calculate the absolute bias and mean squared error (mse) defined as

$$\text{bias} = \frac{1}{nR} \sum_{t=1}^n \sum_{r=1}^R \left| \widehat{g}_j^{(r)}(\tau_t, x_t) - g_j(\tau_t, x_t) \right|, \quad \text{mse} = \frac{1}{nR} \sum_{t=1}^n \sum_{r=1}^R \left(\widehat{g}_j^{(r)}(\tau_t, x_t) - g_j(\tau_t, x_t) \right)^2, \quad (11)$$

to examine the performance of our proposed kernel estimator. We consider the predictive step j to be 1, 2, 3, 6, 9, 12 and the long horizon steps J to be 3, 6, 9, 12. We will report the bias and mse for $\widehat{g}(\tau, x)$ as well. The bias and mse are defined similar to (11). In particular, we consider several different bandwidth selection methods, including the rule-of-thumb, leave-out-out cross-validation method, and the one presented in Appendix A.1. We choose the Gaussian kernel function, i.e., $K(u) = \frac{1}{\sqrt{2\pi}} \exp(-u^2/2)$. The results for $\widehat{g}_j(\tau, x)$ and $\widehat{g}(\tau, x)$ are presented in Tables 1 and 2, respectively.

From Table 1, we can find that (1) the values of absolute bias and mean squared errors of $\widehat{g}_j(\tau, x)$ exhibit a consistent decreasing trend as the sample size increases, which holds true across all three bandwidth selection methods under investigation; (2) notably, our proposed bandwidth selection

Table 1. Simulation results for $\widehat{g}_j(\tau, x)$ with predictive step j being 1, 2, 3, 6, 9 and 12.

	j=1	j=2	j=3	j=6	j=9	j=12	j=1	j=2	j=3	j=6	j=9	j=12
n	Bias						MSE					
Panel A: bandwidth $h_j = \rho_j h_0$												
100	0.4379	0.4247	0.4781	0.4311	0.4219	0.4942	0.3080	0.2950	0.3548	0.3001	0.3006	0.3762
200	0.3975	0.3562	0.3573	0.3573	0.3554	0.3659	0.2501	0.2179	0.2176	0.2173	0.2170	0.2195
400	0.2991	0.2993	0.3006	0.3004	0.2994	0.3000	0.1523	0.1519	0.1538	0.1540	0.1525	0.1530
n	Panel B: Leave-one-out cross-validation											
100	0.4904	0.4862	0.4862	0.4856	0.4818	0.4895	0.3871	0.3815	0.3807	0.3778	0.3741	0.3845
200	0.3906	0.3888	0.3878	0.3925	0.3887	0.3900	0.2587	0.2562	0.2549	0.2586	0.2558	0.2570
400	0.3138	0.3100	0.3104	0.3136	0.3105	0.3122	0.1686	0.1645	0.1658	0.1678	0.1662	0.1666
n	Panel C: Rule-of-thumb											
100	0.4476	0.4477	0.4479	0.4415	0.4457	0.4475	0.3366	0.3376	0.3384	0.3282	0.3345	0.3355
200	0.3785	0.3815	0.3839	0.3808	0.3805	0.3811	0.2483	0.2505	0.2525	0.2479	0.2489	0.2500
400	0.3187	0.3148	0.3147	0.3176	0.3142	0.3164	0.1791	0.1741	0.1751	0.1769	0.1749	0.1761

Panel A illustrates the performance using our proposed bandwidth selection method, as detailed in Appendix A.1. Panels B and C present the corresponding results obtained through leave-one-out cross-validation and rule-of-thumb bandwidth selection methods, respectively.

Table 2. Simulation results for $\widehat{g}(\tau, x)$ with J being 3, 6, 9 and 12.

n	J=3	J=6	J=9	J=12	J=3	J=6	J=9	J=12
	Bias				MSE			
Panel A: bandwidth $h_j = \rho_j h_0$								
100	0.7701	1.2304	1.6764	2.1167	0.9462	2.3779	4.3365	6.8166
200	0.6422	1.0123	1.3420	1.6657	0.6722	1.6388	2.8566	4.3562
400	0.5724	0.9445	1.2978	1.6469	0.5275	1.4090	2.6253	4.1964
Panel B: Leave-one-out cross-validation								
100	0.8011	1.2429	1.6392	2.0202	1.0280	2.4373	4.2012	6.3071
200	0.6495	1.0148	1.3561	1.6817	0.6871	1.6464	2.9032	4.4290
400	0.5723	0.9440	1.2979	1.6463	0.5300	1.4122	2.6333	4.1982
Panel C: Rule-of-thumb								
100	0.8200	1.2198	1.5547	1.8666	1.0869	2.3847	3.8534	5.5453
200	0.7046	1.0628	1.3700	1.6609	0.8157	1.8331	3.0212	4.4188
400	0.5988	0.9177	1.2007	1.4636	0.5953	1.3760	2.3380	3.4567

Panels A, B and C illustrate the performance obtained through our proposed bandwidth selection method, leave-one-out cross-validation and rule-of-thumb bandwidth selection methods, respectively.

method demonstrates superior performance, yielding smaller values of absolute bias and mean squared errors compared to the alternative methods. In addition, the leave-one-out cross-validation and rule-of-thumb bandwidth selection methods show comparable performance in this particular experimental setting.

Table 2 reveals similar findings that (1) for all the three bandwidth selection methods, the values of absolute bias and mean squared errors of $\widehat{g}(\tau, x)$ decrease when sample size increases; (2) our proposed bandwidth selection method performs best in most cases, while rule-of-thumb can outperform when J is relatively large, say 12.

For simplicity, we initially assume a common bandwidth for both scaled time and the random variable x . Following a referee's suggestion, we also investigate the scenario where distinct bandwidths are employed for these two variables. For illustrative purposes, we apply the rule-of-thumb method to select separate bandwidths for each variable. The corresponding results, presented in Table 3, demonstrate comparable performance to those shown in Panel C of Table 1, where a single bandwidth was used. This suggests that while using separate bandwidths offers greater flexibility, the overall estimation accuracy remains similar to the simpler approach under the current experimental setup.

Table 3. Simulation results for $\widehat{g}_j(\tau, x)$ with predictive step j being 1, 2, 3, 6, 9 and 12.

n	j=1	j=2	j=3	j=6	j=9	j=12	j=1	j=2	j=3	j=6	j=9	j=12
	Bias						MSE					
100	0.5321	0.5321	0.5312	0.5310	0.5307	0.5343	0.4233	0.4210	0.4215	0.4181	0.4206	0.4238
200	0.4579	0.4608	0.4580	0.4589	0.4588	0.4594	0.3153	0.3181	0.3174	0.3155	0.3168	0.3174
400	0.3728	0.3686	0.3698	0.3715	0.3702	0.3713	0.2136	0.2089	0.2106	0.2122	0.2107	0.2119

Here we choose different bandwidth for scaled time τ and variable x .

4. Stock return prediction using NPR model

In this section, we employ the NPR model proposed in Section 2 to predict excess stock return or equity premium. We use updated data from Welch and Goyal (2008)⁴ spanning 1965:01–2018:12. The equity premium return is calculated as log return on the S & P 500 minus the log return on a risk-free bill. Following the literature, we consider 14 economic variables as possible predictors, including Log dividend-price ratio ($\log(DP)$), Log dividend yield ($\log(DY)$), Log earnings-price ratio ($\log(EP)$), Log dividend-payout ratio ($\log(DE)$), Stock variance (SVAR), Book-to-market ratio (BM), Net equity expansion (NTIS), Treasury bill rate (TBL), Long-term yield (LTY), Long-term return (LTR), Term spread (TMS), Default yield spread (DFY), Default return spread (DFR), and Inflation (INFL). For definitions on those variables, see Rapach, Strauss, and Zhou (2010) and Rapach and Zhou (2013).

4.1. Evidence for Local Stationarity

We first provide evidence that questions the stationarity of the above commonly used predictors. We consider the trending models $x_{tk} = \mu_k(\tau_t) + \varepsilon_{tk}$ for $k = 1, \dots, 14$, where $\mu_k(\tau)$ is an unknown nonparametric function and $\{\varepsilon_{tk}\}$ is a sequence of locally stationary errors. Figure 1 plots the time-varying means (i.e., $\widehat{\mu}_k(\tau)$) of these 14 predictors with their 95% bootstrap confidence intervals based on the method proposed by Yan, Gao, and Peng (2025). These bootstrap confidence intervals are robust to both local stationarity and serial correlation of error innovations. For the detailed estimation and bootstrap procedures, we refer interested readers to Section 3 of Yan, Gao, and Peng (2025).

Clearly, the estimated trends of these predictors fluctuate over time, suggesting that the stationarity assumption may not hold in practice. Here, we use the bootstrap method to construct the confidence intervals, which remain robust regardless of whether the error innovations are locally stationary. We observe that the length of the confidence intervals varies over time. For instance, the estimation uncertainty for the log DP ratio was high in the 1980s and 2000s, but decreased significantly during the 1980–2000 period. This suggests that our locally stationary assumption for predictors is more reasonable than the stationarity assumption. Note that the stationarity assumption corresponds to a constant asymptotic variance across the entire time period, as stated in Theorem 3.1 of Yan, Gao, and Peng (2025).

4.2. Time-varying/nonlinear predictability

Most of the existing studies use linear regression models to investigate whether stock returns are predictable, and there are also a few exceptions that consider time-varying return predictability based on linear models with time-varying coefficients (Farmer, Schmidt, and Timmermann, 2023; Yu and Yan, 2023) or nonlinear predictability (Zhou et al., 2024) separately. Our framework facilitates the study of both time-varying and nonlinear return predictability. Our first objective is to identify time-varying and

⁴We thank Amit Goyal for providing the data on his web page at <http://www.hec.unil.ch/agoyal/>. In addition, the data used may differ from the real-time information available to market participants, which may be an important consideration for interpreting our results.

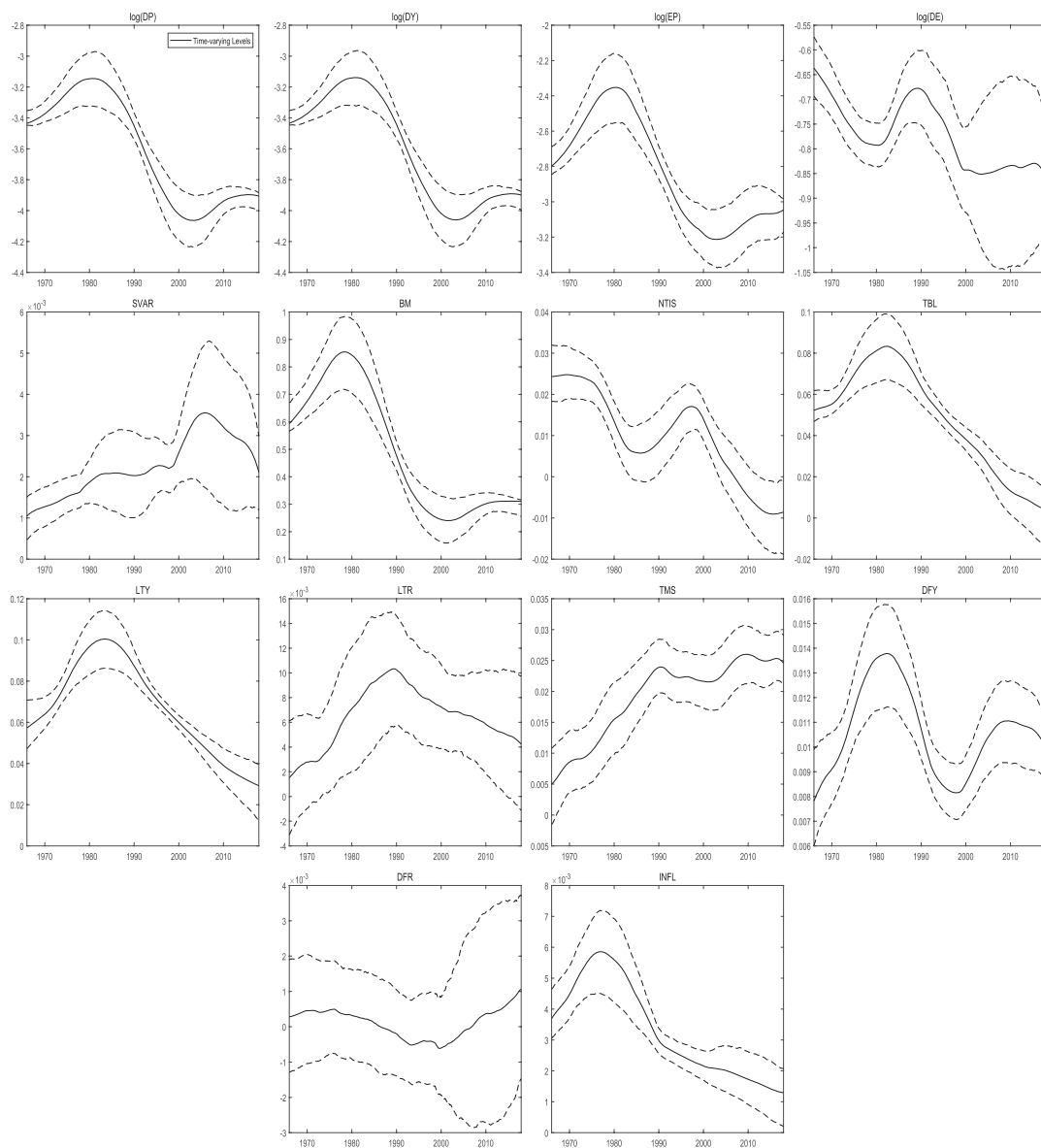


Figure 1. The estimated trend of 14 commonly used predictors as well as the associated 95% bootstrap confidence intervals.

nonlinear patterns of return predictability based on the NPR model, which are of great importance in practice.

We now consider the NPR models $y_{t+1} = g(\tau_t, x_{tk}) + e_{t+1}$ for $k = 1, \dots, 14$, and the univariate linear regression $y_{t+1} = \beta_0 + \beta_1 x_{tk} + e_{t+1}$, $k = 1, \dots, 14$ for comparison. The estimation is done in exactly the same way as in Section 2.2, and we estimate the bandwidth based on the procedures outlined in Appendix A.1.

Figure 2 plots the in-sample patterns of return predictability with respect to time and the value of predictors. Clearly, the return predictability varies over time. We also find strong evidence of nonlinearities in the predictability of stock returns. To precisely visualize the approximate form of nonlinearity, Fig. 3 plots the predicted value of the equity premium, $\hat{g}(\tau, x_{tk})$, against x_{tk} with its

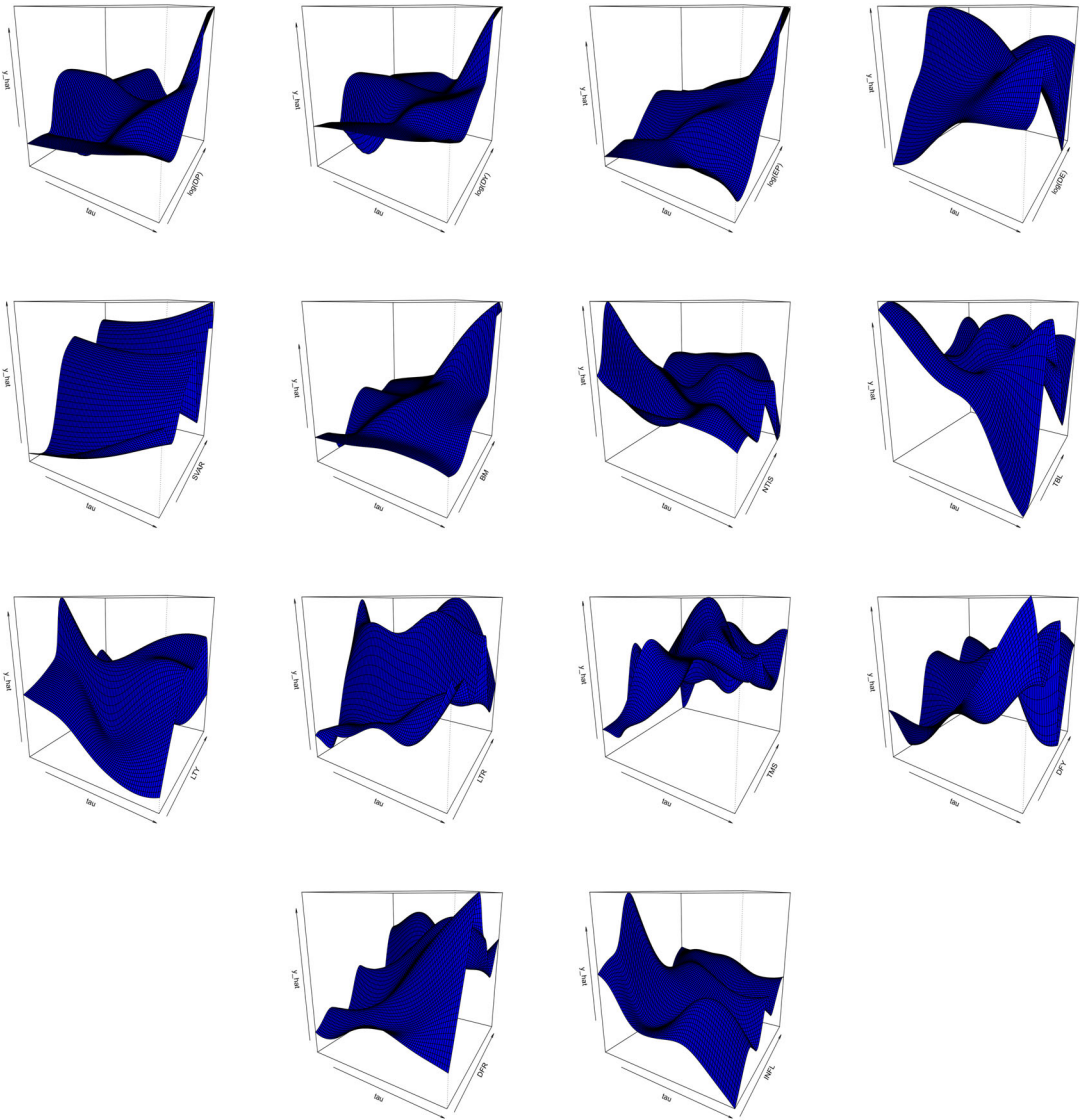


Figure 2. Plots of $\hat{g}(\tau_t, x_{tk})$ with 14 regressors, respectively.

95% confidence intervals for $\tau = 0.8$ and linear regression estimates $\hat{y}_{t+1}$. The confidence intervals are obtained using the asymptotic distribution in [Theorem 2.1](#). [Figure 3](#) shows that these 14 predictors exhibit a nonlinear and complex relationship with stock returns, and that our NPR model captures the nonlinearity in the data better than the linear specification. For example, the two predictors $\log(\text{DE})$ and LTR exhibit a hump-shaped relationship. These results emphasize the importance of accounting for time-varying predictability and nonlinearities when using these predictors to predict stock returns.

4.3. Out-of-sample forecasting performance

In what follows, we evaluate the out-of-sample performance of the NPR model. For comparison purposes, we also consider the commonly used historical mean model and the traditional linear predictive regression model. The former is usually regarded as a benchmark and the latter is also

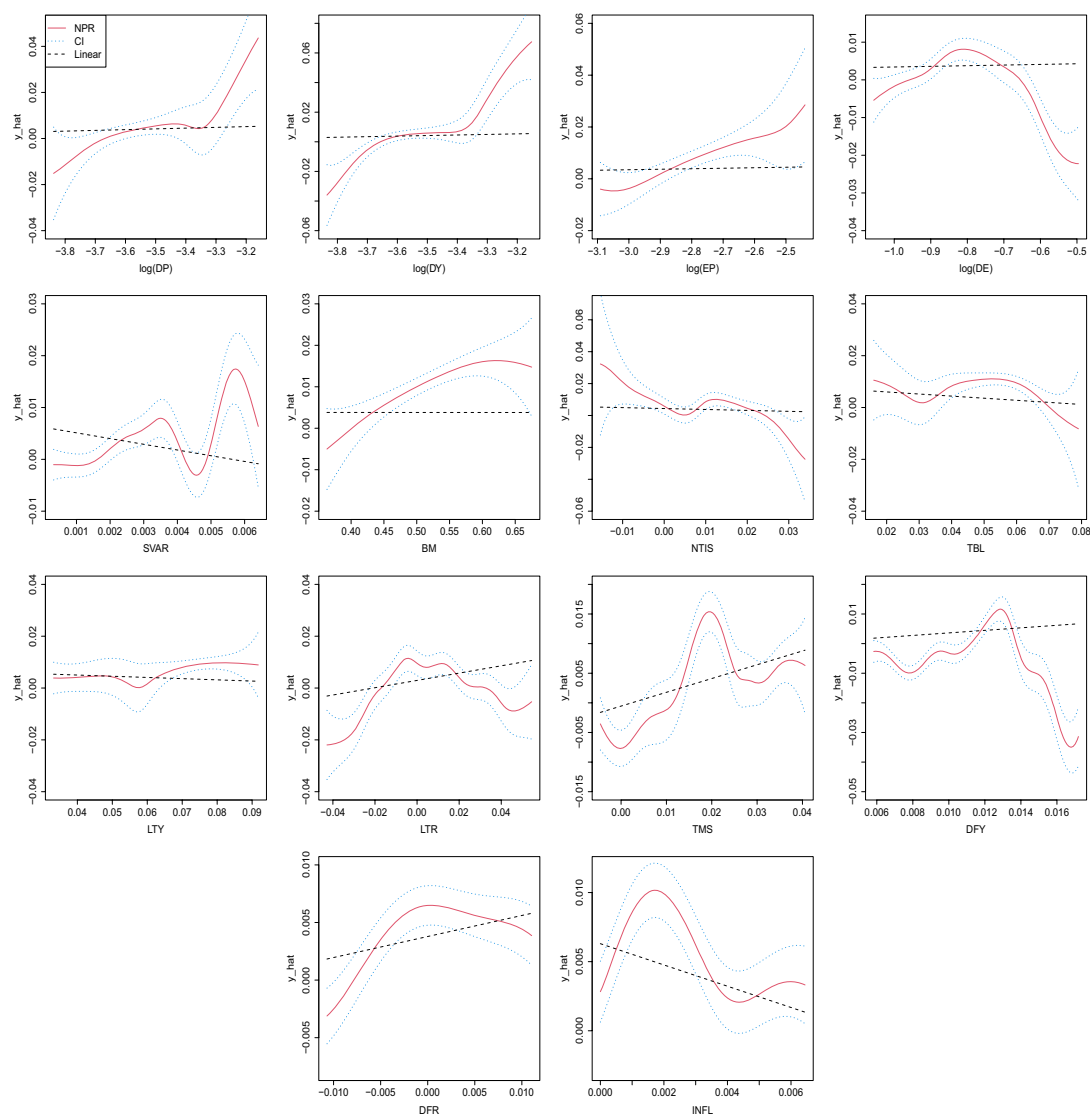


Figure 3. Plots of $\hat{g}(\tau, x_{tk})$ with 95% confidence interval for $\tau = 0.8$ and linear regression estimates $\hat{y}_{t+1}$ with 14 regressors, respectively. In this figure, the red solid lines denote the NPR estimates, blue-dotted lines denote the 95% confidence interval and the black-dashed lines denote the univariate linear estimates.

referred to as a Kitchen Sink linear model, which corresponds to a multiple linear predictive regression model that includes all 14 economic variables as regressors. Moreover, we consider the linear predictive model with time-varying coefficients (TV Model). Following Farmer, Schmidt, and Timmermann (2023), we use the simple equal-weighted average of all 14 univariate models to achieve dimensionality reduction. In addition, as one of the referees suggested, here we add several machine learning methods including LASSO, Ridge, principal component regression (PCR), partial least squares (PLS), iterated mean combination (IMC, Lin, Wu, and Zhou, 2018), iterated weighted combination (IWC, Lin, Wu, and Zhou, 2018), and Bagging algorithm (Jacobsen, Jiang, and Zhang, 2020) for comparison as well.

For NPR model, due to the “curse of dimensionality” problem, we cannot consider all 14 variables simultaneously in one regression, and we include a subset of those variables as shown below. Moreover, in this study, we consider a naive model average approach based on the univariate NPR model, as it

is claimed that in the context of forecasting stock returns, these combination approaches frequently dominate more complicated multivariate models (Rapach and Zhou, 2013). In addition, as pointed out by Chen, Jiang, and Liu (2023), more sophisticated weighting schemes that combine the predictive variable do not consistently perform better. With unstable economic relations and a limited sample size, sophisticated methods may lead to overfitting or be subject to more estimation errors. In such situations, the simple model average method may work better.

To be specific, we consider the following models for stock return prediction.

- Historical mean: $y_{t+j} = \mu + e_{t+j}$;
- Kitchen sink linear model: $y_{t+j} = \alpha_j + \sum_{k=1}^{14} \beta_{jk} x_{tk} + e_{t+j}$; where x_{tk} , $k = 1, 2, \dots, 14$, corresponds to the above 14 predictors.
- NPR multivariate (MV) 1: $y_{t+j} = g_j(\tau_t, x_t^{(1)}) + e_{t+j}$, where $x_t^{(1)} = (\log(DP), \log(DY), TBL)'$;
- NPR multivariate (MV) 2: $y_{t+j} = g_j(\tau_t, x_t^{(2)}) + e_{t+j}$, where $x_t^{(2)} = (BM, TBL, TMS)'$;
- NPR model average (MA) 1: $\hat{y}_{t+j} = \frac{1}{14} \sum_{k=1}^{14} \hat{y}_{t+j,k}$, where $y_{t+j,k} = g_j(\tau_t, x_{tk}) + e_{t+j}$, $k = 1, 2, \dots, 14$;
- NPR model average (MA) 2:

$$\hat{y}_{t+j} = \begin{cases} \frac{1}{6} (\hat{y}_{t+j,1} + \hat{y}_{t+j,2} + \hat{y}_{t+j,3} + \hat{y}_{t+j,8} + \hat{y}_{t+j,9} + \hat{y}_{t+j,10}) & \text{for } 1 \leq j \leq 7 \\ \frac{1}{3} (\hat{y}_{t+j,1} + \hat{y}_{t+j,2} + \hat{y}_{t+j,8}) & \text{for } 8 \leq j \leq 12 \end{cases} \quad (12)$$

where $y_{t+j,k} = g_j(\tau_t, x_{tk}) + e_{t+j}$.

Specifically, for NPR MA1 and NPR MA2, we first compute predictive regression forecasts of the equity premium using only one predictor. Then we obtain a combination forecast by taking the average of those individual forecasts. For NPR MA1, we take simple average of all 14 individual forecasts, while for NPR MA2, we compute simple average for a subset of those 14 forecasts⁵. As Eq. (12) suggests, when the predictive step is smaller than 8, we consider combination forecasts based on individual NPR regressions on respectively, $\log(DP)$, $\log(DY)$, $\log(EP)$, TBL , LTY , and LTR ; and when predictive step is between 8 and 12, we consider regressions with $\log(DP)$, $\log(DY)$, and TBL . In addition, we refer interested readers to Lin, Wu, and Zhou (2018), Jacobsen, Jiang, and Zhang (2020) for a detailed description of the machine learning methods.

We evaluate the out-of-sample performance of these models using the following expansive window scheme, that is, the estimation sample always starts in 1965:01 and additional observations are used as they become available. The details are described as follows.

- For the first window, we conduct the multi-step prediction based on $n - 1$ observations. At the point x_n , we predict y_{n+1} using these $n - 1$ pairs of observations $\{(x_1, y_2), (x_2, y_3), \dots, (x_{n-1}, y_n)\}$. The estimated value of y_{n+1} is denoted as $\hat{y}_{n+1}$. Then we use the observations

$$\{(x_1, y_3), (x_2, y_4), \dots, (x_{n-2}, y_n), (x_{n-1}, \hat{y}_{n+1})\}$$

to predict y_{n+2} at the point x_n . Similarly, we predict y_{n+3} at the point x_n using observations

$$\{(x_1, y_4), (x_2, y_5), \dots, (x_{n-2}, \hat{y}_{n+1}), (x_{n-1}, \hat{y}_{n+2})\}.$$

Repeating such procedure, we obtain the predicted return series for $y_{n+1}, y_{n+2}, \dots, y_{n+J}$ denoted as

$$\hat{y}_{n+1,1}, \hat{y}_{n+1,2}, \dots, \hat{y}_{n+1,J}.$$

- The second window is obtained by expanding the first window to include x_n . At the point x_{n+1} , we conduct the multi-step prediction to predict $y_{n+2}, y_{n+3}, \dots, y_{n+J+1}$ with the predicted values denoted as

$$\hat{y}_{n+2,1}, \hat{y}_{n+2,2}, \dots, \hat{y}_{n+2,J}.$$

⁵This is inspired by empirical study results.

- The procedure continues until we obtain the R th window, that is, the forecast for the last observation in 2018:12 is made. At the point x_{n+R-1} , we conduct the multi-step prediction for $y_{n+R}, y_{n+R+1}, \dots, y_{n+R+J-1}$ and the predicted values are denoted as

$$\hat{y}_{n+R,1}, \hat{y}_{n+R,2}, \dots, \hat{y}_{n+R,J}.$$

We know that the out-of-sample forecast uses only the data available up to the time at which the forecast is made. Therefore, for a given predictive step j , following the work by Campbell and Thompson (2008), we compute the out-of-sample R^2 , which is defined as

$$R_{OOS,j,n,R}^2 = 1 - \frac{\sum_{r=1}^R (y_{n+r,j} - \hat{y}_{n+r,j})^2}{\sum_{r=1}^R (y_{n+r,j} - \bar{y}_{n+r,j})^2},$$

where $\hat{y}_{n+r,j}$ is the j -th step predicted return in the r -th window, $y_{n+r,j}$ is the corresponding observed return, $\bar{y}_{n+r,j}$ is the sample mean of observations using the information up to $n + r - 1$, n is the sample size of the initial data to get a regression estimate at the start of evaluation period, and R is the total number of expansive windows.

As the selection of the out-of-sample (OOS) forecast evaluation period is always somewhat arbitrary, in this study, we consider different OOS periods. Specifically, we consider OOS period: 2012:01-2018:12 and then OOS periods: 1990:01-2018:12, 1990:01-1999:12 and 2000:01-2011:12.

4.3.1. OOS period: 2012:01-2018:12

We first look at the OOS period 2012:01-2018:12, that is, 1965:01-2011:12 are used as the initial in-sample estimation period. The results of $R_{OOS,j,n,R}^2$ with $j = 1, 2, 3, 6, 9, 12$ are presented in Table 4. We can make the following observations.

- For most of the cases, NPR MV1, NPR MV2, NPR MA1, and NPR MA2 can beat both historical mean benchmark, kitchen sink model, machine learning methods, and TV Model in terms of $R_{OOS,j,n,R}^2$ for most cases.
- Among the multivariate NPR and NPR combination forecasts, NPR MV1 performs slightly better than NPR MA2, while the naive model average approach (NPR MA1) has desirable forecasting performance.
- Among the 14 univariate NPR models, the regression with the predictor TBL can beat the historical mean benchmark for all the considered predictive steps. Though 13 of the 14 regressions cannot consistently beat the historical mean benchmark, some of them can frequently beat the benchmark at some predictive steps. For example, the univariate regressions respectively with the following predictors log(DP), log(DY), log(EP), BM, and LTY. After comparing the results of this particular regression with other OOS periods in the following subsections, we find that this phenomenon only happens for this OOS period, so we think this might change when we increase the OOS observations.
- Among those machine learning methods, though PLS, IMC, and IWC have superior performance when $j = 1$, $R_{OOS,j,n,R}^2$ becomes smaller or even negative when j gets larger.
- The TV Model combination forecasts are able to beat historical mean benchmark during 2012–2018 and are quite stable across different forecasting horizons, although the gains are quite moderate.

Overall, the above results suggest that NPR MV1 achieves a balance between predictive accuracy and robustness, and the naive model average approach (NPR MA1) has desirable forecasting performance.

We also examined the out-of-sample prediction for long horizon returns $y_{n:n+J} = \sum_{j=1}^J y_{n+j}$. We define the out-of-sample R^2 as follows: $R_{OOS,J,n,R}^2 = 1 - \frac{\sum_{r=1}^R (y_{n:n+J}^{(r)} - \hat{y}_{n:n+J}^{(r)})^2}{\sum_{r=1}^R (y_{n:n+J}^{(r)} - \sum_{j=1}^J \bar{y}_{n+r,j})^2}$, where $\hat{y}_{n:n+J}^{(r)}$ denotes the estimated value of $y_{n:n+J}^{(r)}$ from the r -th expansive window. With $J = 3, 6, 9, 12$, we present the results of $R_{OOS,J,n,R}^2$ in Table 5, from which we can see that NPR MV1 performs the best, followed by NPR MV2,

Table 4. Results of $R_{OOS,j,n,R}^2$ with $j = 1, 2, 3, 6, 9, 12$ for OOS period: 2012:01-2018:12.

Models	j = 1	j = 2	j = 3	j = 6	j = 9	j = 12
Mean	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Kitchen Sink	0.0875	0.0290	0.0014	0.0128	0.0075	-0.0478
NPR MV1	0.1108	0.0706	0.0474	0.0520	0.0537	0.0185
NPR MV2	0.0628	0.0397	0.0136	0.0609	0.0356	-0.0065
NPR MA1	0.0639	0.0309	0.0270	0.0283	0.0261	-0.0161
NPR MA2	0.1027	0.0504	0.0344	0.0274	0.0379	0.0110
log(DP)	0.0617	0.0449	0.0287	-0.0255	0.0052	0.0096
log(DY)	0.0469	0.0288	0.0220	-0.0042	0.0229	0.0087
log(EP)	0.0558	0.0456	0.0207	0.0266	0.0292	-0.0440
log(DE)	0.0047	-0.0118	0.0048	-0.0139	0.0051	-0.0713
SVAR	0.0250	-0.0046	0.0309	-0.0222	-0.0151	-0.0119
BM	0.0225	0.0312	0.0372	0.0600	0.0502	-0.0087
NTIS	0.0250	0.0257	0.0160	0.0348	0.0019	-0.2200
TBL	0.0683	0.0595	0.0495	0.0699	0.0623	0.0101
LTY	0.0697	0.0468	0.0243	0.0616	0.0240	-0.0223
LTR	0.0941	-0.0283	-0.0785	-0.1196	0.0021	-0.0384
TMS	0.0270	0.0108	0.0186	-0.0473	-0.0194	-0.0381
DFY	0.0154	-0.0260	0.0034	-0.0922	-0.0559	0.0035
DFR	0.0311	-0.1361	-0.2075	-0.1169	-0.1458	-0.0802
INFL	-0.0567	-0.0719	-0.0208	0.0117	0.0418	-0.0219
LASSO	-0.0007	-0.0003	-0.0001	-0.0027	-0.0024	-0.0044
Ridge	-0.0059	-0.0130	-0.0174	-0.0226	-0.0155	-0.0231
PCR	-0.0067	-0.0145	-0.0176	-0.0232	-0.0122	-0.0164
PLS	0.0396	0.0300	-0.0248	0.0096	-0.0003	-0.0455
IMC	0.0110	-0.0005	-0.0051	-0.0024	-0.0051	-0.0099
IWC	0.0110	-0.0005	-0.0051	-0.0024	-0.0051	-0.0099
Bagging	-0.0827	-0.0669	-0.0614	0.0359	-0.0415	-0.0065
TV Model	0.0124	0.0066	0.0034	0.0086	0.0096	0.0126

Table 5. Results of $R_{OS,j,n,R}^2$ for OOS period: 2012:01-2018:12.

Models	J = 3	J = 6	J = 9	J = 12
Mean	0.00000	0.00000	0.00000	0.00000
Kitchen Sink	0.18471	0.18884	0.22670	0.21553
NPR MV1	0.26347	0.45401	0.50575	0.49650
NPR MV2	0.15587	0.34182	0.40728	0.36237
NPR MA1	0.13974	0.28611	0.30951	0.23915
NPR MA2	0.18710	0.33016	0.36213	0.35233
LASSO	-0.00587	-0.02017	-0.02523	-0.03273
Ridge	-0.06879	-0.18476	-0.21871	-0.23343
PCR	-0.06703	-0.18826	-0.21564	-0.21998
PLS	0.16681	0.20739	0.16084	0.12560
IMC	0.01179	-0.01235	-0.03221	-0.04936
IWC	0.01187	-0.01231	-0.03208	-0.04922
Bagging	0.02844	0.21172	0.28464	0.18297
TV Model	0.00755	0.00746	0.00750	0.00887

NPR MA2, and then NPR MA1. In addition, PLS, Bagging and TV Model have superior performance compared to the historical mean model for all considered horizons. And the mean model is not a good choice in these cases. Overall, for this OOS period, multivariate NPR and NPR combination forecasts perform better than historical mean, kitchen sink regression, machine learning methods, and TV model.

4.3.2. OOS period: 1990:01-2018:12

In this section, we evaluate the out-of-sample performance of NPR model for the OOS period: 1990:01-2018:12. There are several interesting observations. As we can see from Table 6, historical mean model performs best in almost all the predictive steps for this OOS period, but if we remove the forecasts for the year 2009, NPR MA1 and NPR MA2 can beat the historical mean when the predictive step j is small

Table 6. Results of $R^2_{OOS,j,n,R}$ with $j = 1, 2, 3, 6, 9, 12$ for OOS period: 1990:01-2018:12.

OOS Period	1990-2018						1990-2008, 2010-2018 (without 2009)					
Models	j=1	j=2	j=3	j=6	j=9	j=12	j=1	j=2	j=3	j=6	j=9	j=12
Mean	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Kitchen	-0.1133	-0.1899	-0.1745	-0.2066	-0.1539	-0.1256	-0.0288	-0.1044	-0.1094	-0.1204	-0.0846	-0.0780
NPR MV1	-0.0991	-0.1438	-0.0520	-0.2152	-0.1785	-0.0913	-0.0532	-0.0442	0.0192	-0.1670	-0.1447	-0.0750
NPR MV2	-0.0699	-0.0673	-0.0559	-0.0594	-0.0386	-0.0651	-0.0592	-0.0486	-0.0274	-0.0492	-0.0285	-0.0663
NPR MA1	-0.0144	-0.0068	-0.0054	-0.0245	-0.0226	-0.0226	-0.0006	0.0166	0.0160	-0.0218	-0.0178	-0.0180
NPR MA2	0.0081	-0.0221	-0.0064	-0.0242	-0.0213	-0.0078	0.0176	0.0070	0.0208	-0.0181	-0.0139	-0.0057
log(DP)	-0.0573	-0.0513	-0.0342	-0.0470	-0.0324	-0.0153	-0.0030	0.0169	0.0312	-0.0352	-0.0270	-0.0130
log(DY)	-0.0535	-0.0606	-0.0253	-0.0468	-0.0338	-0.0106	-0.0146	0.0090	0.0269	-0.0354	-0.0231	-0.0073
log(EP)	-0.0189	-0.0189	-0.0319	-0.0206	-0.0306	-0.0361	-0.0023	0.0019	0.0027	-0.0135	-0.0217	-0.0315
log(DE)	-0.1266	-0.0335	-0.0483	-0.0583	-0.0904	-0.0540	-0.0573	0.0126	0.0011	-0.0525	-0.0839	-0.0467
SVAR	-0.1744	-0.1010	-0.0581	-0.0456	-0.0355	-0.0431	-0.1050	-0.0615	-0.0296	-0.0449	-0.0316	-0.0328
BM	-0.0165	-0.0123	-0.0082	0.0049	-0.0071	-0.0207	-0.0087	-0.0003	0.0060	0.0107	-0.0056	-0.0189
NTIS	-0.0546	-0.0401	-0.0032	-0.0533	-0.0456	-0.0710	-0.0267	-0.0130	0.0128	-0.0437	-0.0344	-0.0524
TBL	-0.0309	-0.0266	-0.0194	-0.0401	-0.0144	-0.0103	-0.0237	-0.0108	-0.0022	-0.0311	-0.0079	-0.0094
LTY	-0.0102	-0.0056	-0.0060	-0.0180	-0.0244	-0.0332	-0.0026	0.0056	0.0154	-0.0077	-0.0170	-0.0300
LTR	-0.0082	-0.0828	-0.0425	-0.0600	-0.0604	-0.0500	0.0091	-0.0636	-0.0390	-0.0554	-0.0517	-0.0468
TMS	-0.0443	-0.0377	-0.0300	-0.0445	-0.0212	-0.0326	-0.0401	-0.0333	-0.0098	-0.0338	-0.0153	-0.0332
DFY	-0.1444	-0.0234	-0.0612	-0.0526	-0.0478	-0.0339	-0.0088	0.0571	0.0113	-0.0413	-0.0524	-0.0256
DFR	-0.1651	-0.1290	-0.0918	-0.1858	-0.0744	-0.0905	-0.0986	-0.0828	-0.0558	-0.1349	-0.0626	-0.0660
INFL	-0.0800	-0.0412	-0.0439	-0.0867	-0.0646	-0.0737	-0.0953	-0.0307	-0.0335	-0.0834	-0.0571	-0.0600
LASSO	-0.0033	-0.0044	-0.0029	-0.0056	-0.0023	-0.0032	-0.0003	-0.0010	-0.0011	-0.0041	-0.0027	-0.0040
Ridge	-0.0278	-0.0347	-0.0319	-0.0373	-0.0261	-0.0262	-0.0127	-0.0168	-0.0187	-0.0300	-0.0260	-0.0293
PCR	-0.0549	-0.0622	-0.0614	-0.0732	-0.0456	-0.0469	-0.0322	-0.0356	-0.0383	-0.0617	-0.0445	-0.0528
PLS	-0.0809	-0.1457	-0.1512	-0.1598	-0.1104	-0.0944	-0.0337	-0.0823	-0.0879	-0.0917	-0.0674	-0.0955
IMC	-0.0095	-0.0184	-0.0128	-0.0152	-0.0105	-0.0106	-0.0056	-0.0143	-0.0121	-0.0143	-0.0110	-0.0127
IWC	-0.0095	-0.0184	-0.0128	-0.0153	-0.0105	-0.0106	-0.0056	-0.0143	-0.0121	-0.0144	-0.0110	-0.0128
Bagging	-0.2636	-0.3222	-0.2000	-0.2503	-0.1757	-0.1674	-0.2339	-0.3281	-0.2081	-0.1554	-0.0822	-0.1249
TV Model	-0.0009	-0.0023	-0.0023	-0.0013	-0.0001	-0.0008	0.0052	0.0013	0.0005	0.0005	0.0004	0.0011

and TV Model can beat the historical mean for all horizons. In addition, $R^2_{OOS,j,n,R}$ for the kitchen sink regression, all the 14 univariate NPR regressions and those machine learning methods become larger after removing the prediction for the year 2009. We also observe that multivariate NPR (MV1 and MV2) and those machine learning methods do not have much advantage compared with conventional forecasts for this OOS period.

For the prediction of long horizon returns with results presented in Tables 7, we can find that if we consider the OOS period 1990:01-2018:12, the historical mean model has absolute advantage in terms of $R^2_{OS,j,n,R}$, while after removing the forecasts in 2009, NPR MA1 can beat historical mean benchmark when $J \leq 6$, NPR MA2 can beat historical mean benchmark when $J \leq 9$ and TV Model can beat the historical mean for all horizons. In addition, in all cases, kitchen sink model and machine learning methods perform very poorly.

4.3.3. OOS period: 1990:01-1999:12

We now look at the OOS period 1990:01-1999:12, that is, 1965:01-1989:12 are used as the initial in-sample estimation period. We present the results of $R^2_{OOS,j,n,R}$ with $j = 1, 2, 3, 6, 9, 12$ in Table 8, from which we can make the following observations.

- For most of the cases, NPR MV2, NPR MA1, and NPR MA2 can beat both historical mean benchmark, kitchen sink model, machine learning methods, and TV model in terms of $R^2_{OOS,j,n,R}$.
- Model combination forecasts (NPR MA1 and MA2) perform better than multivariate NPR.
- Among the 14 univariate NPR models, the regressions with predictors BM, LTY, TMS, and DFY can beat the historical mean benchmark for all the considered predictive steps. Though the rest 10 univariate regressions cannot consistently beat the historical mean benchmark, most of them can

Table 7. Results of $R_{OS,jn,R}^2$ for OOS period: 1990:01-2018:12.

OOS Period	1990-2018				1990-2008, 2010-2018 (without 2009)			
Models	J=3	J=6	J=9	J=12	J=3	J=6	J=9	J=12
Mean	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Kitchen Sink	-0.43101	-0.76991	-1.06156	-1.23951	-0.21753	-0.41995	-0.55562	-0.62678
NPR MV1	-0.25377	-0.56082	-0.83001	-1.02742	-0.05724	-0.29296	-0.54971	-0.71630
NPR MV2	-0.16765	-0.26681	-0.33863	-0.40587	-0.11411	-0.19944	-0.26076	-0.34450
NPR MA1	-0.01745	-0.05789	-0.09526	-0.14399	0.04036	0.02312	-0.00948	-0.05573
NPR MA2	-0.02171	-0.04788	-0.07701	-0.09002	0.04410	0.03591	0.00463	-0.01273
LASSO	-0.01448	-0.03081	-0.04121	-0.05119	-0.00664	-0.02475	-0.03989	-0.05329
Ridge	-0.09714	-0.18974	-0.26084	-0.30483	-0.04677	-0.13098	-0.20428	-0.26438
PCR	-0.17790	-0.35437	-0.48165	-0.55959	-0.10070	-0.25915	-0.38480	-0.48088
PLS	-0.34651	-0.63977	-0.88534	-1.04790	-0.18090	-0.36330	-0.47919	-0.56612
IMC	-0.04696	-0.08135	-0.10930	-0.13362	-0.03936	-0.07617	-0.10552	-0.13283
IWC	-0.04708	-0.08160	-0.10959	-0.13394	-0.03943	-0.07638	-0.10578	-0.13311
Bagging	-0.51502	-0.79633	-1.00673	-1.23816	-0.45280	-0.71991	-0.93592	-1.16730
TV Model	-0.00160	-0.00108	-0.00092	-0.00046	0.00245	0.00231	0.00158	0.00148

Table 8. Results of $R_{OS,jn,R}^2$ with $j = 1, 2, 3, 6, 9, 12$ for OOS period: 1990:01-1999:12.

Models	j = 1	j = 2	j = 3	j = 6	j = 9	j = 12
Mean	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Kitchen Sink	-0.1300	-0.1831	-0.1694	-0.2543	-0.1844	-0.1869
NPR MV1	-0.0642	0.0276	-0.0259	-0.0171	0.0378	0.0318
NPR MV2	-0.0181	0.0084	-0.0067	0.0308	0.0432	0.0278
NPR MA1	0.0063	0.0221	0.0098	0.0344	0.0397	0.0558
NPR MA2	0.0054	0.0146	0.0161	0.0358	0.0302	0.0501
log(DP)	-0.0211	0.0208	0.0131	0.0361	0.0280	0.0490
log(DY)	-0.0431	0.0002	-0.0026	0.0384	0.0176	0.0422
log(EP)	-0.0343	-0.0138	-0.0108	0.0409	0.0446	0.0574
log(DE)	-0.0976	0.0066	-0.0165	0.0164	0.0237	0.0561
SVAR	0.0270	0.0141	-0.0022	0.0159	0.0382	0.0588
BM	0.0270	0.0300	0.0217	0.0466	0.0322	0.0451
NTIS	-0.0534	0.0089	-0.0205	0.0135	0.0409	0.0462
TBL	-0.0001	0.0270	0.0218	0.0414	0.0397	0.0548
LTY	0.0007	0.0066	0.0113	0.0411	0.0382	0.0455
LTR	0.0306	-0.0114	0.0029	-0.0230	-0.0235	0.0636
TMS	0.0074	0.0167	0.0174	0.0464	0.0470	0.0419
DFY	0.0349	0.0313	0.0248	0.0424	0.0364	0.0412
DFR	-0.0444	0.0096	-0.0409	0.0058	0.0457	0.0810
INFL	-0.0067	0.0024	-0.0370	0.0247	0.0346	0.0211
LASSO	-0.0025	-0.0019	-0.0009	-0.0040	-0.0008	-0.0032
Ridge	-0.0677	-0.0610	-0.0559	-0.0731	-0.0488	-0.0571
PCR	-0.1533	-0.1313	-0.1225	-0.1623	-0.1015	-0.1162
PLS	-0.1081	-0.1058	-0.1229	-0.1762	-0.0578	-0.0884
IMC	-0.0200	-0.0206	-0.0225	-0.0316	-0.0182	-0.0261
IWC	-0.0199	-0.0206	-0.0225	-0.0318	-0.0182	-0.0262
Bagging	-0.4437	-0.8412	-0.5983	-0.4905	-0.1741	-0.2977
TV Model	0.0351	0.0348	0.0327	0.0304	0.0351	0.0359

frequently beat the benchmark at some predictive steps. For example, the univariate regressions, respectively, with the following predictors $\log(DP)$, $\log(DY)$, $\log(EP)$, $\log(DE)$, and $NTIS$.

For the prediction of long horizon returns with results presented in Table 9, we can find that NPR MA1 performs best in terms of $R_{OS,jn,R}^2$, followed by NPR MA2, NPR MV2, TV model, and then NPR MV1. Among all the cases, kitchen sink model and machine learning methods perform very poorly in this OOS period. Overall, for this OOS period, NPR combination forecasts perform better than historical mean, kitchen sink regression, machine learning methods and TV model.

Table 9. Results of $R_{OS,Jn,R}^2$ for OOS period: 1990:01-1999:12.

Models	J = 3	J = 6	J = 9	J = 12
Mean	0.00000	0.00000	0.00000	0.00000
Kitchen	-0.47371	-1.23263	-1.97278	-2.30382
NPR MV1	-0.05427	-0.14038	0.06076	0.08809
NPR MV2	-0.00175	0.06308	0.24233	0.34737
NPR MA1	0.04892	0.17756	0.33252	0.43515
NPR MA2	0.03713	0.15785	0.32176	0.42408
LASSO	-0.01109	-0.03492	-0.05706	-0.07657
Ridge	-0.20699	-0.48106	-0.75601	-0.88555
PCR	-0.45008	-1.05127	-1.61883	-1.85113
PLS	-0.34668	-0.94518	-1.42255	-1.62411
IMC	-0.07354	-0.20968	-0.33315	-0.41323
IWC	-0.07355	-0.21024	-0.33401	-0.41425
Bagging	-1.50651	-2.65643	-3.43680	-3.80152
TV Model	0.03433	0.03220	0.03326	0.03361

4.3.4. OOS period: 2000:01-2011:12

In this section, we evaluate the out-of-sample performance of NPR model for the OOS period: 2000:01-2011:12. Overall, the performance of all the considered models in this OOS period is very similar to those in 1990:01-2018:12, which means that the performance of those models in this particular period dominate the overall performance in 1990:01-2018:12, since we already show that the OOS performance of NPR models in both 1990:01-1999:12 and 2012:01-2018:12 are better than conventional linear forecasts, historical mean benchmark, and machine learning methods.

Specifically, as we can see from Table 10, historical mean model performs best in almost all the predictive steps for this OOS period, but if we remove the forecasts for the year 2009, NPR MA1, NPR MA2, Ridge, and PCR can beat the historical mean when the predictive step j is small. In addition, $R_{OS,j,n,R}^2$ for the kitchen sink regression, all the 14 univariate NPR regressions and those machine learning methods become larger after removing the prediction for the year 2009. We also observe that multivariate NPR (MV1 and MV2) do not have much advantage compared with conventional forecasts for this OOS period.

For the prediction of long horizon returns with results presented in Table 11, we can find that if we consider the OOS period 2000:01-2011:12, the historical mean model has absolute advantage in terms of both $R_{OS,Jn,R}^2$, while after removing the forecasts in 2009, NPR MA1, MA2, and Bagging can beat historical mean benchmark when $J = 3$, and Ridge and PCR can beat historical mean benchmark when $J = 3, 6$. In addition, in all cases, kitchen sink model performs very poorly.

In summary, we can see that the multivariate NPR and NPR combination forecasts can significantly outperform the historical mean benchmark, traditional linear predictive regression models, machine learning methods, and TV model in certain cases in terms of out-of-sample forecasting. This is particularly true for OOS periods that do not include the year 2009, where multivariate NPR and NPR combination forecasts clearly outperform the historical mean benchmark. Overall, it is found that the out-of-sample forecasting performance of all considered methods varies significantly over time, indicating time-varying return predictability. Additionally, NPR combination forecasts exhibit more stable and robust performance for different out-of-sample periods, while the naive model average approach has desirable forecasting performance, overcomes the curse of dimensionality, and avoids complex model selection procedures.

4.4. Trading strategy

We can see that NPR model and combined forecasts can sometimes beat the historical mean model according to out-of-sample prediction measured by out-of-sample R^2 . We now turn to an economic metric for comparing these methods with the historical mean model. Specifically, we propose an explicit

Table 10. Results of $R^2_{OOS,j,n,R}$ with $j = 1, 2, 3, 6, 9, 12$ for OOS period: 2000:01–2011:12.

OOS Period	2000–2011						2000–2008, 2010–2011 (without 2009)					
Models	j=1	j=2	j=3	j=6	j=9	j=12	j=1	j=2	j=3	j=6	j=9	j=12
Mean	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Kitchen	−0.1641	−0.2618	−0.2385	−0.2645	−0.1950	−0.1310	−0.0011	−0.1007	−0.1163	−0.0978	−0.0657	−0.0366
NPR MV1	−0.1605	−0.2933	−0.0835	−0.3971	−0.3807	−0.1909	−0.0783	−0.1161	0.0544	−0.3172	−0.3365	−0.1701
NPR MV2	−0.1217	−0.1257	−0.0988	−0.1138	−0.1166	−0.1155	−0.1096	−0.0973	−0.0470	−0.0982	−0.1051	−0.1240
NPR MA1	−0.0403	−0.0281	−0.0159	−0.0621	−0.0659	−0.0492	−0.0169	0.0158	0.0254	−0.0601	−0.0610	−0.0431
NPR MA2	−0.0120	−0.0522	−0.0229	−0.0634	−0.0654	−0.0261	0.0035	0.0017	0.0293	−0.0546	−0.0555	−0.0240
log(DP)	−0.1023	−0.1137	−0.0699	−0.0913	−0.0720	−0.0362	0.0002	0.0140	0.0560	−0.0718	−0.0657	−0.0340
log(DY)	−0.0782	−0.1167	−0.0416	−0.0936	−0.0734	−0.0240	−0.0035	0.0150	0.0602	−0.0749	−0.0571	−0.0191
log(DE)	−0.0157	−0.0320	−0.0545	−0.0528	−0.0865	−0.0654	0.0188	0.0079	0.0116	−0.0416	−0.0750	−0.0598
log(DE)	−0.1771	−0.0563	−0.0709	−0.1075	−0.1781	−0.0892	−0.0451	0.0332	0.0246	−0.1001	−0.1748	−0.0788
SVAR	−0.3537	−0.1738	−0.0951	−0.0887	−0.0664	−0.0892	−0.2456	−0.1055	−0.0427	−0.0912	−0.0621	−0.0741
BM	−0.0479	−0.0443	−0.0295	−0.0186	−0.0449	−0.0415	−0.0378	−0.0251	−0.0036	−0.0092	−0.0459	−0.0403
NTIS	−0.0650	−0.0749	0.0127	−0.0980	−0.1101	−0.0846	−0.0100	−0.0258	0.0461	−0.0827	−0.0951	−0.0491
TBL	−0.0658	−0.0707	−0.0491	−0.0999	−0.0702	−0.0353	−0.0577	−0.0459	−0.0182	−0.0873	−0.0634	−0.0365
LTY	−0.0253	−0.0245	−0.0252	−0.0623	−0.0711	−0.0594	−0.0127	−0.0049	0.0151	−0.0457	−0.0617	−0.0561
LTR	−0.0903	−0.0946	−0.0493	−0.0832	−0.0876	−0.0964	−0.0703	−0.0577	−0.0431	−0.0762	−0.0735	−0.0954
TMS	−0.0782	−0.0536	−0.0501	−0.0851	−0.0700	−0.0570	−0.0759	−0.0472	−0.0123	−0.0675	−0.0637	−0.0610
DFY	−0.2898	−0.0570	−0.1262	−0.1016	−0.0865	−0.0642	−0.0402	0.1003	0.0106	−0.0832	−0.0995	−0.0512
DFR	−0.2153	−0.2373	−0.1126	−0.2930	−0.0964	−0.1554	−0.0889	−0.1613	−0.0433	−0.2009	−0.0759	−0.1141
INFL	−0.1299	−0.0578	−0.0568	−0.1524	−0.1394	−0.1207	−0.1704	−0.0392	−0.0376	−0.1517	−0.1329	−0.0988
LASSO	−0.0045	−0.0069	−0.0045	−0.0071	−0.0030	−0.0027	0.0012	−0.0007	−0.0012	−0.0044	−0.0037	−0.0042
Ridge	−0.0110	−0.0248	−0.0222	−0.0199	−0.0157	−0.0094	0.0207	0.0108	0.0053	−0.0032	−0.0136	−0.0129
PCR	−0.0119	−0.0344	−0.0386	−0.0338	−0.0220	−0.0143	0.0381	0.0204	0.0102	−0.0056	−0.0159	−0.0207
PLS	−0.0962	−0.2089	−0.1880	−0.1902	−0.1617	−0.1110	−0.0075	−0.0964	−0.0713	−0.0638	−0.0856	−0.1154
IMC	−0.0089	−0.0205	−0.0087	−0.0086	−0.0064	−0.0017	−0.0011	−0.0134	−0.0062	−0.0059	−0.0065	−0.0046
IWC	−0.0090	−0.0206	−0.0087	−0.0087	−0.0064	−0.0017	−0.0012	−0.0135	−0.0063	−0.0059	−0.0065	−0.0046
Bagging	−0.1543	−0.0621	0.0575	−0.2184	−0.2772	−0.1334	−0.0967	−0.0216	0.0659	−0.0449	−0.1161	−0.0714
TV Model	−0.0251	−0.0260	−0.0241	−0.0221	−0.0229	−0.0225	−0.0171	−0.0230	−0.0221	−0.0219	−0.0252	−0.0256

Table 11. Results of $R^2_{OS,j,n,R}$ for OOS period: 2000:01–2011:12.

OOS Period	2000–2011				2000–2008, 2010–2011 (without 2009)			
Models	J=3	J=6	J=9	J=12	J=3	J=6	J=9	J=12
Mean	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Kitchen Sink	−0.52865	−0.77284	−0.99871	−1.17340	−0.16377	−0.22696	−0.23032	−0.21312
NPR MV1	−0.42003	−0.82205	−1.26926	−1.63679	−0.09520	−0.44090	−0.90312	−1.21498
NPR MV2	−0.28309	−0.42228	−0.58096	−0.73809	−0.20480	−0.33913	−0.49435	−0.67658
NPR MA1	−0.06401	−0.16664	−0.26358	−0.36450	0.03187	−0.05558	−0.15580	−0.24971
NPR MA2	−0.07473	−0.15742	−0.24653	−0.30645	0.03442	−0.04212	−0.14512	−0.20831
LASSO	−0.01793	−0.03194	−0.03893	−0.04623	−0.00184	−0.01487	−0.02555	−0.03509
Ridge	−0.06005	−0.10459	−0.12460	−0.12670	0.03303	0.00159	−0.02476	−0.04734
PCR	−0.08435	−0.16101	−0.19203	−0.20381	0.06163	0.01511	−0.01696	−0.05086
PLS	−0.43632	−0.67111	−0.89927	−1.08151	−0.16826	−0.24592	−0.30476	−0.34823
IMC	−0.04641	−0.05344	−0.05944	−0.06092	−0.03054	−0.03386	−0.03918	−0.04258
IWC	−0.04660	−0.05364	−0.05964	−0.06110	−0.03066	−0.03400	−0.03933	−0.04272
Bagging	−0.07891	−0.27030	−0.47319	−0.65676	0.02754	−0.13227	−0.36052	−0.55040
TV Model	−0.02472	−0.02312	−0.02282	−0.02254	−0.02039	−0.02011	−0.02138	−0.02205

trading strategy that switches between stocks and bonds based on whether predicted stock returns are greater than a threshold. We also compare this strategy with the buy and hold strategy that just holds stocks for the duration.

We first employ our proposed NPR model (including the multivariate NPR and combination forecasts) to predict stock returns respectively, and obtain their corresponding one-step-ahead forecasts, then we compare these values with a chosen threshold. If the corresponding value is greater than the given threshold, we put money in stock market; Otherwise we buy a risk-free bond. So our trading strategy earns in one period is $w_t r_{t+1} + (1 - w_t) r_{t+1}^0$, where r_{t+1}^0 is the t-bill rate next period and r_{t+1}

Table 12. Trading strategy with the use of NPR MV1.

Starting date	Profit under different thresholds								Buy & hold
	0.0014	0.0024	0.0034	0.0042	0.0052	0.0057	0.0067	0.0079	
1990:01	3.0546	3.4652	4.0218	3.7858	3.3401	2.8219	2.8922	2.7470	6.6178
2000:01	0.4955	0.8337	1.2011	0.9681	0.9547	0.8778	0.5875	0.8229	0.7977
2010:01	1.0456	1.1623	1.1623	0.9885	0.8668	0.7633	0.6863	0.7676	1.3344
Turnover ratios under different thresholds									
1990:01	0.1925	0.2557	0.3190	0.3563	0.4023	0.4224	0.4828	0.5287	–
2000:01	0.2325	0.3070	0.3596	0.3772	0.3947	0.4079	0.4561	0.5044	–
2010:01	0.0185	0.0278	0.0278	0.0370	0.0463	0.0556	0.0833	0.1389	–
Sharpe ratio with no transaction cost									
1990:01	0.1384	0.1522	0.1696	0.1680	0.1624	0.1511	0.1576	0.1585	0.1583
2000:01	0.0724	0.1064	0.1405	0.1249	0.1264	0.1208	0.0934	0.1208	0.0772
2010:01	0.2063	0.2236	0.2236	0.2054	0.1903	0.1760	0.1639	0.1810	0.2283
Sharpe ratio with 25bps transaction cost									
1990:01	0.1239	0.1320	0.1431	0.1377	0.1272	0.1137	0.1132	0.1080	0.1579
2000:01	0.0537	0.0799	0.1076	0.0895	0.0884	0.0810	0.0480	0.0689	0.0766
2010:01	0.2049	0.2216	0.2216	0.2027	0.1868	0.1718	0.1575	0.1702	0.2270

Table 13. Trading strategy with the use of NPR MV2.

Starting date	Profit under different thresholds								Buy & hold
	0.0014	0.0024	0.0034	0.0042	0.0052	0.0057	0.0067	0.0079	
1990:01	3.9930	3.7620	5.0058	5.6963	5.3168	6.2217	6.9263	5.2873	6.6178
2000:01	0.7312	0.6688	1.0204	1.1632	1.1742	1.2341	1.8412	1.0068	0.7977
2010:01	0.7207	0.5961	0.5520	0.6452	0.6984	0.7003	1.0585	0.7647	1.3344
Turnover ratios under different thresholds									
1990:01	0.2414	0.2730	0.3017	0.3448	0.3764	0.3966	0.4310	0.4885	–
2000:01	0.2939	0.3246	0.3509	0.3860	0.4079	0.4254	0.4561	0.5175	–
2010:01	0.0833	0.1111	0.1204	0.1759	0.1852	0.1944	0.2222	0.2778	–
Sharpe ratio with no transaction cost									
1990:01	0.1577	0.1573	0.1863	0.2023	0.1975	0.2130	0.2387	0.2280	0.1583
2000:01	0.0950	0.0908	0.1270	0.1437	0.1455	0.1512	0.2099	0.1604	0.0772
2010:01	0.1763	0.1548	0.1467	0.1755	0.1871	0.1874	0.2743	0.2309	0.2283
Sharpe ratio with 25bps transaction cost									
1990:01	0.1393	0.1359	0.1613	0.1727	0.1650	0.1784	0.1980	0.1783	0.1579
2000:01	0.0707	0.0635	0.0948	0.1065	0.1059	0.1095	0.1606	0.0969	0.0766
2010:01	0.1696	0.1458	0.1370	0.1602	0.1708	0.1703	0.2526	0.2021	0.2270

is the outturn on the stock market next period and our weights are $w_t = I(\hat{r}_{t+1} > c)$, in which c is a selected threshold. In this study, we consider eight different thresholds, which are corresponding to quantiles between 40% and 60% of historical distribution of returns, to examine the performance of our trading strategy with the buy and hold strategy in terms of profit. We compute the profit of both strategies based on the S & P 500 Index and assume that the cost such as transaction fee during the trading could be ignored. To check the robustness of our proposed trading strategy, we consider three investment starting dates, which are 1990:01, 2000:01, and 2010:01. We also calculate the monthly Sharpe ratio of the portfolio, which is the mean portfolio return in excess of the risk-free rate divided by the standard deviation of the excess portfolio return. To improve the calculations of actual investment returns, we also add 25bps transaction costs in the calculations of Sharpe ratio.

Tables 12–15 show the results of stock return predictions that with multivariate NPR and combination models, respectively. In each table, we present the profit, turnover ratios and Sharpe ratios with or without transaction costs of the trading strategy under different thresholds in each case. From these results, we can see that NPR MA2 is more robust as there always exists some thresholds under which our proposed strategies can outperform the buy and hold strategy in terms of profit no matter when the investment started. For example, for the NPR MA2, the thresholds are 0.0034, 0.0042 and 0.0057. As a result, we see that our proposed trading strategies with the use of NPR combination forecasts could be better alternatives to the buy and hold strategy. When comparing the Sharpe ratios of the trading

Table 14. Trading strategy with the use of NPR MA1.

Starting date	Profit under different thresholds								Buy & hold
	0.0014	0.0024	0.0034	0.0042	0.0052	0.0057	0.0067	0.0079	
1990:01	4.8660	5.3438	4.0856	4.9928	4.9080	5.4948	4.7473	3.5800	6.6178
2000:01	0.7057	0.9249	0.5489	0.8192	0.9080	0.9584	0.7074	0.4689	0.7977
2010:01	1.0456	1.1445	1.0214	1.0579	1.0635	0.9548	0.4600	0.2551	1.3344
Turnover ratios under different thresholds									
1990:01	0.1609	0.2069	0.2787	0.3506	0.4425	0.4828	0.5661	0.6293	–
2000:01	0.2237	0.2807	0.3596	0.4254	0.4912	0.5132	0.6053	0.6798	–
2010:01	0.0185	0.0463	0.0741	0.1019	0.1481	0.1759	0.2870	0.4167	–
Sharpe ratio with no transaction cost									
1990:01	0.1677	0.1780	0.1613	0.1863	0.1963	0.2100	0.2142	0.2013	0.1583
2000:01	0.0900	0.1102	0.0796	0.1083	0.1235	0.1303	0.1226	0.1033	0.0772
2010:01	0.2063	0.2255	0.2099	0.2164	0.2281	0.2133	0.1443	0.1072	0.2283
Sharpe ratio with 25bps transaction cost									
1990:01	0.1558	0.1623	0.1397	0.1572	0.1569	0.1661	0.1575	0.1332	0.1579
2000:01	0.0723	0.0870	0.0493	0.0701	0.0756	0.0791	0.0504	0.0082	0.0766
2010:01	0.2049	0.2221	0.2044	0.2088	0.2164	0.1993	0.1175	0.0596	0.2270

Table 15. Trading strategy with the use of NPR MA2.

Starting date	Profit under different thresholds								Buy & hold
	0.0014	0.0024	0.0034	0.0042	0.0052	0.0057	0.0067	0.0079	
1990:01	7.1750	6.9419	6.9158	8.6317	6.4012	6.7625	4.9322	6.1245	6.6178
2000:01	1.0673	1.0083	1.1568	1.5898	1.5028	1.7458	1.3886	2.1150	0.7977
2010:01	1.2004	1.2269	1.4359	1.4153	1.3519	1.4425	1.1832	1.5009	1.3344
Turnover ratios under different thresholds									
1990:01	0.1408	0.1724	0.2328	0.3103	0.4368	0.5000	0.5690	0.6552	–
2000:01	0.1842	0.2325	0.2895	0.3553	0.4649	0.5219	0.5921	0.6754	–
2010:01	0.0463	0.0741	0.1111	0.1204	0.1944	0.2222	0.2500	0.3981	–
Sharpe ratio with no transaction cost									
1990:01	0.1946	0.1972	0.2026	0.2292	0.2188	0.2311	0.2095	0.2539	0.1583
2000:01	0.1148	0.1153	0.1303	0.1667	0.1740	0.1965	0.1793	0.2566	0.0772
2010:01	0.2353	0.2413	0.2783	0.2757	0.2795	0.2991	0.2681	0.3459	0.2283
Sharpe ratio with 25bps transaction cost									
1990:01	0.1843	0.1841	0.1844	0.2038	0.1800	0.1850	0.1550	0.1838	0.1579
2000:01	0.1005	0.0963	0.1055	0.1342	0.1274	0.1422	0.1143	0.1732	0.0766
2010:01	0.2318	0.2357	0.2695	0.2662	0.2633	0.2801	0.2463	0.3070	0.2270

strategy, we also find that NPR MA2 performs best. Specifically, NPR MA2 outperforms the the buy and hold strategy regardless of the threshold used and regardless of when the investment was started, after adjusting for the potential risk of the trading strategy and transaction costs.

5. Conclusions

In this article, we have introduced the multi-step NPR model, which allows for general locally stationary predictors and time-varying/nonlinear return predictability. Estimation theory and asymptotic properties have been established in both the short and long horizons. Moreover, we have employed the NPR model to investigate monthly stock return predictability over the period 1965–2018. The empirical results show that the multivariate NPR and NPR combination forecasts can substantially outperform the historical mean benchmark, traditional linear predictive regression model, and several machine learning methods for some cases in terms of out-of-sample forecasting performance. Overall, we have found that (1) the out-of-sample forecasting performance of all considered methods varies significantly over time, indicating time-varying return predictability and (2) the performance of NPR combination forecasts is more stable and robust for different out-of-sample periods and there indeed exist evidence of stock return predictability in some out-of-sample periods, though it is noteworthy that the events of 2009 had a great effect on the forecasting performance of NPR models. It is too early to tell whether the recent

Covid-19 pandemic would similarly affect stock return predictability, although some newspaper articles have argued that so far momentum strategies have tended to do quite well during this crisis, see Fletcher and Wigglesworth (2020). In addition, we have also compared our methods with the historical mean benchmark according to an economic metric. We have showed how our method can be used to define a trading strategy that beats the buy and hold strategy over our sample period.

Disclosure statement

The authors would like to declare that there is no conflict of interest.

Data availability statement

The datasets involved in this paper will be made available to the reader upon to the acceptance of this submission.

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A.

In this appendix, we first discuss computational details on the implementation of the NPR model, particularly the bandwidth selection and the estimation of asymptotic variance, and then present the proofs of the main results for the estimators in the NPR models.

A.1. Bandwidth selection

As we mentioned in Section 2, we use the local constant kernel method to estimate the unknown function $g_j(\cdot)$ in the NPR model. It is generally accepted that the performance of the kernel estimator is mainly determined by bandwidth. In the last 30 years, there has been a comprehensive list of studies on the bandwidth selection. This appendix focuses on the issue of how to choose ρ_j and h involved in $h_j = \rho_j h$ used in the estimation of model (1).

Our approach is motivated by existing studies in Härdle, Hall, and Marron (1988), Härdle, Hall, and Marron (1989), Fan and Gijbels (1995), Xia and Li (2002), and Cheng, Gao, and Zhang (2019). Let us introduce the following notation:

$$D_j(h_j) = \frac{1}{n} \sum_{t=1}^n \left(\widehat{g}_j(\tau_t, x_t) - g_j(\tau_t, x_t) \right)^2 w(\tau_t, x_t), \quad (\text{A1})$$

where $w(\cdot, \cdot)$ is a probability kernel function satisfying $\int_{-\infty}^{\infty} \int_0^1 w^2(\tau, u) d\tau du < \infty$. Let $\widehat{h}_j$ be chosen such that it minimizes $D_j(h_j)$ over all possible $\{h_j\}$ by using a leave-one-out cross-validation method (Xia and Li, 2002; Richter and Dahlhaus, 2019). Let h_{j0} be chosen such that it minimizes $d_j(h_j) = E[D_j(h_j)]$. In view of both the establishment and the proofs of the results in Xia and Li (2002), it can be shown that as $n \rightarrow \infty$,

$$n^{\frac{3}{10}} \left(\frac{\widehat{h}_j}{h_{j0}} - 1 \right) \rightarrow_D N(0, \sigma_{j0}^2) \quad (\text{A2})$$

for each fixed j , where $0 < \min_{j \geq 1} \sigma_{j0}^2 \leq \max_{j \geq 1} \sigma_{j0}^2 < \infty$, and $h_{j0} = \rho_j h_0$ with $\rho_j = j^\beta$ or θ^j , in which $h_0 > 0$, $\beta > 0$ and $\theta > 1$ will all be estimated in the rest of this section.

Using Eq. (A2), we have for large enough n ,

$$\log \left(\frac{\widehat{h}_j}{h_{j0}} \right) = \log \left(1 + \frac{\widehat{h}_j}{h_{j0}} - 1 \right) \approx \frac{\widehat{h}_j}{h_{j0}} - 1 \equiv n^{-\frac{3}{10}} \varepsilon_j, \quad (\text{A3})$$

where $\varepsilon_j = n^{\frac{3}{10}} \left(\frac{\widehat{h}_j}{h_{j0}} - 1 \right) \rightarrow_D N(0, \sigma_{j0}^2)$. This suggests an approximate regression model of the form

$$\begin{aligned} \log(\widehat{h}_j) &= \log(h_{j0}) + \eta_j = \log(h_0) + \log(\rho_j) + \eta_j \\ &= \begin{cases} \log(h_0) + \beta \log(j) + \eta_j, & \text{if } \rho_j = j^\beta, \\ \log(h_0) + j \log(\theta) + \eta_j, & \text{if } \rho_j = \theta^j, \end{cases} \end{aligned} \quad (\text{A4})$$

where $\eta_j = n^{-\frac{3}{10}} \varepsilon_j$ can be viewed as a sequence of random errors with $E[\eta_j] = 0$ and $0 < E[\eta_j^2] = n^{-\frac{3}{5}} \sigma_{j0}^2$.

We then focus on the case of either $\rho_j = j^\beta$ or $\rho_j = \theta^j$. Let $Z_j = \log(\widehat{h}_j)$. For the case of $\rho_j = j^\beta$, we can estimate β by an ordinary least squares (OLS) estimator of the form

$$\widehat{\beta} = \left(\sum_{j=1}^J (\log(j) - \overline{\log(J)})^2 \right)^{-1} \sum_{j=1}^J (\log(j) - \overline{\log(J)}) (Z_j - \overline{Z}), \quad (\text{A5})$$

where $\overline{\log(J)} = \frac{1}{J} \sum_{j=1}^J \log(j)$ and $\overline{Z} = \frac{1}{J} \sum_{j=1}^J Z_j$.

Equations (A4) and (A5) imply that the following rate of convergence:

$$\widehat{\beta} - \beta = O_P \left(\left(\sqrt{J \log(J)} \right)^{-1} \cdot n^{-\frac{3}{10}} \right). \quad (\text{A6})$$

For the case of $\rho_j = \theta^j$, the OLS estimator of $\gamma = \log(\theta)$ is given by

$$\hat{\gamma} = \left(\sum_{j=1}^J (j - \bar{j})^2 \right)^{-1} \sum_{j=1}^J (j - \bar{j}) (Z_j - \bar{Z}), \quad (\text{A7})$$

where $\bar{j} = \frac{1}{J} \sum_{j=1}^J j = \frac{(J+1)}{2}$.

Meanwhile, Eqs. (A4) and (A7) imply a rate of convergence of the form:

$$\hat{\gamma} - \gamma = O_p \left(J^{-\frac{3}{2}} \cdot n^{-\frac{3}{10}} \right). \quad (\text{A8})$$

We finally estimate h_0 by $\hat{h}_0 = \frac{1}{J} \sum_{j=1}^J \hat{h}_j \hat{\rho}_j^{-1}$, where $\hat{\rho}_j = j^{\hat{\beta}}$ or $\hat{\theta}^j$, in which $\hat{\theta} = e^{\hat{\gamma}}$.

Equations (A6) and (A8) imply that the OLS estimators may have fast convergence rates. If we do choose $h_0 = n^{-\frac{1}{5}}$ and assume that $h_j \rightarrow 0$ as $(n, j) \rightarrow (\infty, \infty)$, there will be some restrictions on (J, n) such that either $j^{\hat{\beta}} \cdot n^{-\frac{1}{5}} \rightarrow 0$ or $\hat{\theta}^j \cdot n^{-\frac{1}{5}} \rightarrow 0$ as $(n, J) \rightarrow (\infty, \infty)$.

A step-by-step implementation for bandwidth selection is given as follows.

Step 1: Compute the leave-one-out estimator. For each t , compute the leave-one-out estimator $\hat{g}_j^{(-t)}(\tau_t, x_t)$ using the local constant kernel method:

$$\hat{g}_j^{(-t)}(\tau_t, x_t) = \sum_{s \neq t} W_{ns}(\tau_t, x_t; h_j) y_{s+j},$$

where:

$$W_{ns}(\tau_t, x_t; h_j) = \frac{K\left(\frac{\tau_s - \tau_t}{h_j}\right) \prod_{i=1}^d K\left(\frac{x_s^i - x_t^i}{h_j}\right)}{\sum_{r \neq t} K\left(\frac{\tau_r - \tau_t}{h_j}\right) \prod_{i=1}^d K\left(\frac{x_r^i - x_t^i}{h_j}\right)}.$$

Step 2: Implement leave-one-out cross-validation. The goal is to minimize the cross-validation criterion $D_j(h_j)$ in (A1). Find the bandwidth h_j that minimizes $D_j(h_j)$. This can be done using a grid search or an optimization algorithm.

Step 3: Estimate ρ_j and h . Once h_j is obtained for each j , estimate ρ_j and h using the following regression model:

$$\log(\hat{h}_j) = \log(h_0) + \log(\rho_j) + \eta_j,$$

where η_j is the error term, ρ_j can be modeled as $\rho_j = j^{\beta}$ or $\rho_j = \theta^j$. For $\rho_j = j^{\beta}$, estimate β by (A5). For $\rho_j = \theta^j$, estimate $\gamma = \log(\theta)$ by (A7).

Step 4: Estimate h_0 . We estimate h_0 as:

$$\hat{h}_0 = \frac{1}{J} \sum_{j=1}^J \hat{h}_j \hat{\rho}_j^{-1},$$

where $\hat{\rho}_j = j^{\hat{\beta}}$ or $\hat{\theta}^j$, depending on the chosen model for ρ_j .

Step 5: Implement the bandwidth selection. Use the estimated $\hat{h}_0$ and $\hat{\rho}_j$ to compute the final bandwidth $h_j = \hat{\rho}_j \hat{h}_0$ for each j .

A.2. Estimation of asymptotic variance

Theorems 2.1 and 2.2 provide the asymptotic variances for $\hat{g}_j(\tau, x)$ and $\hat{g}(\tau, x)$, respectively. To construct estimators for these asymptotic variances, we need to estimate the following components.

1. **Conditional variance $\sigma_j^2(x)$:** We can estimate the conditional variance $\sigma_j^2(x)$ using the residuals $e_{t+j} = y_{t+j} - \hat{g}_j(\tau_t, x_t)$. A kernel estimator for $\sigma_j^2(x)$ is:

$$\hat{\sigma}^2(x) = \frac{\sum_{t=1}^n K\left(\frac{\tau_t - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right) e_{t+j}^2}{\sum_{t=1}^n K\left(\frac{\tau_t - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right)}.$$

2. **Marginal density** $f(\tau, x)$: We can estimate the marginal density $f(\tau, x)$ using a kernel density estimator:

$$\widehat{f}(\tau, x) = \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K\left(\frac{\tau_t - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right).$$

3. **Bandwidth** h_j : The bandwidth h_j can be selected using our proposed method as described in Appendix A.1 of the article or other bandwidth selection methods (say, leave-one-out cross-validation).

A.3. Proofs of the main theorems

Proof of Theorem 2.1. Observe that

$$\widehat{g}_j(\tau, x) - g_j(\tau, x) = \frac{1}{\widehat{f}(\tau, x)} \left(\widehat{g}_j^E(\tau, x) + \widehat{g}_j^B(\tau, x) - g_j(\tau, x) \widehat{f}(\tau, x) \right), \quad (\text{A9})$$

where we let $L(x) = \prod_{i=1}^d K(x^i)$ and then write

$$\begin{aligned} \widehat{f}(\tau, x) &= \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K\left(\frac{\tau_t - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right), \\ \widehat{g}_j^E(\tau, x) &= \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K\left(\frac{\tau_t - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right) e_{t+j}, \\ \widehat{g}_j^B(\tau, x) &= \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K\left(\frac{\tau_t - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right) g_j(\tau_t, x_t). \end{aligned}$$

Let $B_j(\tau, x) = \sqrt{nh_j^{d+1}} \left(\widehat{g}_j^B(\tau, x) - g_j(\tau, x) \widehat{f}(\tau, x) \right)$ denote the bias part and $V_j(\tau, x) = \sqrt{nh_j^{d+1}} \widehat{g}_j^E(\tau, x)$ denote the stochastic part.

Then we have $(\widehat{g}_j(\tau, x) - g_j(\tau, x)) = (nh_j^{d+1})^{-1/2} \widehat{f}(\tau, x)^{-1} (V_j(\tau, x) + B_j(\tau, x))$.

We then proceed with the following three steps to show the asymptotic normality of the estimator $\widehat{g}_j(\tau, x)$. The steps are similar to the proof of Theorem 4.3 in Vogt (2012).

- We first show that $B_j(\tau, x) = \sqrt{nh_j^{d+1}} f(\tau, x) \left(B_{j,\tau,x} + o_P(h_j^2) \right)$, where $B_{j,\tau,x} = h_j^2 (R_j(\tau, x) + b_j(\tau, x))$.
- We establish the asymptotic normality $V_j(\tau, x) \rightarrow_D N(0, \kappa_0^{d+1} \sigma_f^2(x) f(\tau, x))$, where $\kappa_0 = \int K^2(u) du$.
- We then show that $\widehat{f}(\tau, x) - f(\tau, x) = o_P(1)$ and $\widehat{f}(\tau, x)^{-1} = O_P(1)$.

Following the spirit of Vogt (2012) that approximate the locally stationary time series x_t by its stationary counterpart $x_t(\tau_t)$, we write

$$\mathbb{E}[\widehat{g}_j^B(\tau, x) - g_j(\tau, x) \widehat{f}(\tau, x)] = Q_1(\tau, x) + \cdots + Q_4(\tau, x),$$

where $Q_i(\tau, x) = \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K_h(\tau - \tau_t) q_i(\tau, x)$, $K_h(x) = K(x/h)$ and

$$\begin{aligned} q_1(\tau, x) &= \mathbb{E} \left[\prod_{i=1}^d \bar{K}_h(x^i - x_t^i) \left\{ \prod_{i=1}^d K_h(x^i - x_t^i) - \prod_{i=1}^d K_h(x^i - x_t^i(\tau_t)) \right\} \times \left\{ g_j(\tau_t, x_t) - g_j(\tau, x) \right\} \right] \\ q_2(\tau, x) &= \mathbb{E} \left[\prod_{i=1}^d \bar{K}_h(x^i - x_t^i) \prod_{i=1}^d K_h(x^i - x_t^i(\tau_t)) \times \left\{ g_j(\tau_t, x_t) - g_j(\tau_t, x_t(\tau_t)) \right\} \right], \\ q_3(\tau, x) &= \mathbb{E} \left[\left\{ \prod_{i=1}^d \bar{K}_h(x^i - x_t^i) - \prod_{i=1}^d \bar{K}_h(x^i - x_t^i(\tau_t)) \right\} \right] \end{aligned}$$

$$\times \prod_{i=1}^d K_h(x^i - x_t^i(\tau_t)) \left\{ g_j(\tau_t, x_t(\tau_t)) - g_j(\tau, x) \right\},$$

$$q_4(\tau, x) = \mathbb{E} \left[\prod_{i=1}^d K_h(x^i - x_t^i(\tau_t)) \left\{ g_j(\tau_t, x_t(\tau_t)) - g_j(\tau, x) \right\} \right],$$

in which $\bar{K}$ is a Lipschitz continuous function with support $[-qC_1, qC_1]$ for some $q > 1$ and $\bar{K}(x) = 1$ for all $x \in [-C_1, C_1]$ and write $\bar{K}_h(x) = \bar{K}(x/h)$.

As the kernel function is bounded, we have

$$\left| \prod_{i=1}^d K_h(x^i - x_t^i) - \prod_{i=1}^d K_h(x^i - x_t^i(\tau_t)) \right| \leq C \sum_{k=1}^d \left| K_h(x^k - x_t^k) - K_h(x^k - x_t^k(\tau_t)) \right|^r,$$

where C is a finite constant and $r = \min\{\omega, 1\}$. Then we have

$$|Q_1(\tau, x)| \leq \frac{C}{nh_j^{d+1}} \sum_{t=1}^n K_h(\tau - \tau_t) \times \mathbb{E} \left[\sum_{k=1}^d \left| K_h(x^k - x_t^k) - K_h(x^k - x_t^k(\tau_t)) \right|^r \right. \\ \left. \times \prod_{i=1}^d \bar{K}_h(x^i - x_t^i) \left| g_j(\tau_t, x_t) - g_j(\tau, x) \right| \right].$$

Under Assumptions A.1.1(i) and A.1.4(i), we further have

$$|Q_1(\tau, x)| \leq \frac{C}{nh_j^d} \sum_{t=1}^n K_h(\tau - \tau_t) \mathbb{E} \left[\sum_{k=1}^d \left| K_h(x^k - x_t^k) - K_h(x^k - x_t^k(\tau_t)) \right|^r \right] \\ \leq \frac{C}{nh_j^d} \sum_{t=1}^n K_h(\tau - \tau_t) \mathbb{E} \left[\sum_{k=1}^d \left| \frac{1}{nh_j} U_{nt}(\tau_t) \right|^r \right] \leq \frac{C}{n^r h_j^{d-1+r}}.$$

Similarly, we can show that $|Q_2(\tau, x)| \leq \frac{C}{n^r h_j^d}$ and $|Q_3(\tau, x)| \leq \frac{C}{n^r h_j^{d-1+r}}$. These results are uniformly in τ and x .

Define

$$\hat{f}^*(\tau, x) = \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K \left(\frac{\tau_t - \tau}{h_j} \right) L \left(\frac{x_t(\tau_t) - x}{h_j} \right),$$

$$\hat{g}_j^{B*}(\tau, x) = \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K \left(\frac{\tau_t - \tau}{h_j} \right) L \left(\frac{x_t(\tau_t) - x}{h_j} \right) g_j(\tau_t, x_t(\tau_t)).$$

Then we can write $Q_4(\tau, x) = \mathbb{E} \left[\hat{g}_j^{B*}(\tau, x) - g_j(\tau, x) \hat{f}^*(\tau, x) \right]$.

Under Assumptions A.1.1(i), A.1.2(ii)(iii), A.1.4(i) and by change of variables, Taylor expansion, we can show that

$$\mathbb{E} \hat{f}^*(\tau, x) = \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K \left(\frac{\tau_t - \tau}{h_j} \right) \mathbb{E} \left[L \left(\frac{x_t(\tau_t) - x}{h_j} \right) \right] \\ = \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K \left(\frac{\tau_t - \tau}{h_j} \right) \left[\int L \left(\frac{x_y - x}{h_j} \right) f(\tau_t, x_y) dx_y \right] \\ = \frac{1 + o_P(1)}{h_j^{d+1}} \iint K \left(\frac{\tau_y - \tau}{h_j} \right) L \left(\frac{x_y - x}{h_j} \right) f(\tau_y, x_y) d\tau_y dx_y \\ = \frac{1 + o_P(1)}{h_j^{d+1}} \int \cdots \int K \left(\frac{\tau_y - \tau}{h_j} \right) \prod_{i=1}^d K \left(\frac{x_y^i - x}{h_j} \right) f(\tau_y, x_y^1, \dots, x_y^d) d\tau_y dx_y^1 \cdots dx_y^d$$

$$\begin{aligned}
&= \int \cdots \int K(p) \prod_{i=1}^d K(q_i) f(\tau + ph_j, x^1 + q_1 h_j, \dots, x^d + q_d h_j) dp dq^1 \cdots dq^d (1 + o_P(1)) \\
&= \int \cdots \int K(p) \prod_{i=1}^d K(q_i) f(\tau, x) dp dq^1 \cdots dq^d \\
&\quad + \int \cdots \int K(p) \prod_{i=1}^d K(q_i) \left(ph_j \frac{\partial f(\tau, x)}{\partial \tau} + \sum_{i=1}^d q_i h_j \frac{\partial f(\tau, x)}{\partial x^i} \right) dp dq^1 \cdots dq^d \\
&\quad + \int \cdots \int K(p) \prod_{i=1}^d K(q_i) \frac{1}{2} \left(p^2 h_j^2 \frac{\partial^2 f(\tau, x)}{\partial \tau^2} + \sum_{i=1}^d q_i^2 h_j^2 \frac{\partial^2 f(\tau, x)}{\partial x^{i2}} \right) dp dq^1 \cdots dq^d \\
&\quad + \int \cdots \int K(p) \prod_{i=1}^d K(q_i) \frac{1}{2} \left(\sum_{i=1}^d p q_i h_j^2 \frac{\partial^2 f(\tau, x)}{\partial \tau \partial x^i} + 2 \sum_{i=2}^d \sum_{s=1}^{i-1} q_i q_s h_j^2 \frac{\partial^2 f(\tau, x)}{\partial x^i \partial x^s} \right) dp dq^1 \cdots dq^d \\
&\quad + o_P(h_j^2) = f(\tau, x) + \frac{\kappa_2}{2} h_j^2 \left(\frac{\partial^2 f(\tau, x)}{\partial \tau^2} + \sum_{i=1}^d \frac{\partial^2 f(\tau, x)}{\partial x^{i2}} \right) + o_P(h_j^2). \tag{A10}
\end{aligned}$$

Similarly, we can show that

$$\begin{aligned}
\mathbb{E}[\widehat{g}_j^{B*}(\tau, x)] &= \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K\left(\frac{\tau_t - \tau}{h_j}\right) \mathbb{E}\left[L\left(\frac{x_t(\tau_t) - x}{h_j}\right) g_j(\tau_t, x_t(\tau_t))\right] \\
&= \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K\left(\frac{\tau_t - \tau}{h_j}\right) \left[\int L\left(\frac{x_y - x}{h_j}\right) g_j(\tau_t, x_y) f(\tau_t, x_y) dx_y \right] \\
&= \frac{1 + o_P(1)}{h_j^{d+1}} \iint K\left(\frac{\tau_y - \tau}{h_j}\right) L\left(\frac{x_y - x}{h_j}\right) g_j(\tau_y, x_y) f(\tau_y, x_y) d\tau_y dx_y \\
&= \frac{1 + o_P(1)}{h_j^{d+1}} \int \cdots \int K\left(\frac{\tau_y - \tau}{h_j}\right) \prod_{i=1}^d K\left(\frac{x_y^i - x}{h_j}\right) \\
&\quad g_j(\tau_y, x_y^1, \dots, x_y^d) f(\tau_y, x_y^1, \dots, x_y^d) d\tau_y dx_y^1 \cdots dx_y^d.
\end{aligned}$$

Taking the second-order Taylor expansion for $g_j(\tau_y, x_y^1, \dots, x_y^d)$ and $f(\tau_y, x_y^1, \dots, x_y^d)$ and keeping the terms up to $O_P(h_j^2)$, we obtain that

$$\begin{aligned}
\mathbb{E}[\widehat{g}_j^{B*}(\tau, x)] &= g_j(\tau, x) f(\tau, x) + \frac{\kappa_2}{2} h_j^2 \left(2 \frac{\partial g_j(\tau, x)}{\partial \tau} \frac{\partial f(\tau, x)}{\partial \tau} + \frac{\partial^2 g_j(\tau, x)}{\partial \tau^2} f(\tau, x) \right) \\
&\quad + \frac{\kappa_2}{2} h_j^2 \sum_{i=1}^d \left(2 \frac{\partial g_j(\tau, x)}{\partial x^i} \frac{\partial f(\tau, x)}{\partial x^i} + \frac{\partial^2 g_j(\tau, x)}{\partial x^{i2}} f(\tau, x) \right) \\
&\quad + \frac{\kappa_2}{2} h_j^2 \left(\frac{\partial^2 f(\tau, x)}{\partial \tau^2} g_j(\tau, x) + \sum_{i=1}^d \frac{\partial^2 f(\tau, x)}{\partial x^{i2}} g_j(\tau, x) \right) + o_P(h_j^2).
\end{aligned}$$

Then we have

$$\begin{aligned}
Q_4(\tau, x) &= \mathbb{E}[\widehat{g}_j^{B*}(\tau, x)] - g_j(\tau, x) \mathbb{E}[\widehat{f}^*(\tau, x)] \\
&= \frac{\kappa_2}{2} h_j^2 \sum_{i=1}^d \left(2 \frac{\partial g_j(\tau, x)}{\partial x^i} \frac{\partial f(\tau, x)}{\partial x^i} + \frac{\partial^2 g_j(\tau, x)}{\partial x^{i2}} f(\tau, x) \right) \\
&\quad + \frac{\kappa_2}{2} h_j^2 \left(2 \frac{\partial g_j(\tau, x)}{\partial \tau} \frac{\partial f(\tau, x)}{\partial \tau} + \frac{\partial^2 g_j(\tau, x)}{\partial \tau^2} f(\tau, x) + o_P(h_j^2) \right)
\end{aligned}$$

$$\begin{aligned}
&= f(\tau, x) \left(h_j^2 R_j(\tau, x) + h_j^2 b_j(\tau, x) + o_P(h_j^2) \right) \\
&= f(\tau, x) \left(B_{j,\tau,x} + o_P(h_j^2) \right),
\end{aligned} \tag{A11}$$

where $B_{j,\tau,x} = h_j^2 R_j(\tau, x) + h_j^2 b_j(\tau, x)$ and

$$\begin{aligned}
R_j(\tau, x) &= \frac{\kappa_2}{2} \sum_{i=1}^d \left(2 \frac{\partial g_j(\tau, x)}{\partial x^i} \frac{\partial f(\tau, x)}{\partial x^i} + \frac{\partial^2 g_j(\tau, x)}{\partial x^{i^2}} f(\tau, x) \right) / f(\tau, x), \\
b_j(\tau, x) &= \frac{\kappa_2}{2} \left(2 \frac{\partial g_j(\tau, x)}{\partial \tau} \frac{\partial f(\tau, x)}{\partial \tau} + \frac{\partial^2 g_j(\tau, x)}{\partial \tau^2} f(\tau, x) \right) / f(\tau, x).
\end{aligned}$$

Define $B_j^*(\tau, x) = \frac{1}{\sqrt{nh_j^{d+1}}} B_j(\tau, x)$. Then we have

$$\mathbb{E}[B_j^*(\tau, x)] = Q_1(\tau, x) + \cdots + Q_4(\tau, x).$$

Combining the results of $Q_1(\tau, x), \dots, Q_4(\tau, x)$, we have that

$$\mathbb{E}[B_j^*(\tau, x)] = f(\tau, x) B_{j,\tau,x} + o_P(h_j^2) + O_P\left(\frac{1}{n^r h_j^d}\right). \tag{A12}$$

As we assume that $n^r h_j^{d+2} \rightarrow \infty$, we have

$$\mathbb{E}[B_j^*(\tau, x)] - f(\tau, x) B_{j,\tau,x} = o_P(h_j^2) = o_P(1).$$

Similar to the derivation of Eq. (A12), we can show that

$$\text{Var}[B_j^*(\tau, x)] = \mathbb{E}[B_j^{*2}(\tau, x)] - \left(\mathbb{E}[B_j^*(\tau, x)] \right)^2 = o_P(1). \tag{A13}$$

Hence, we have $B_j^*(\tau, x) - \mathbb{E}[B_j^*(\tau, x)] = o_P(1)$.

Therefore, we have

$$B_j^*(\tau, x) - f(\tau, x) B_{j,\tau,x} = B_j^*(\tau, x) - \mathbb{E}[B_j^*(\tau, x)] + \mathbb{E}[B_j^*(\tau, x)] - f(\tau, x) B_{j,\tau,x} = o_P(1),$$

which is equivalent as

$$\frac{B_j(\tau, x)}{\sqrt{nh_j^{d+1}}} f(\tau, x)^{-1} - B_{j,\tau,x} = o_P(1). \tag{A14}$$

On the other hand, we have that

$$V_j(\tau, x) = \sqrt{nh_j^{d+1}} \hat{g}_j^E(\tau, x) = \frac{1}{\sqrt{nh_j^{d+1}}} \sum_{t=1}^n K\left(\frac{\tau_t - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right) e_{t+j}.$$

It is obvious that $\mathbb{E}[V_j(\tau, x)] = 0$ and

$$\begin{aligned}
V_j^2(\tau, x) &= \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K^2\left(\frac{\tau_t - \tau}{h_j}\right) L^2\left(\frac{x_t - x}{h_j}\right) e_{t+j}^2 \\
&\quad + \frac{2}{nh_j^{d+1}} \sum_{t=2}^n \sum_{s=1}^{t-1} K\left(\frac{\tau_t - \tau}{h_j}\right) K\left(\frac{\tau_s - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right) L\left(\frac{x_s - x}{h_j}\right) e_{t+j} e_{s+j}, \\
&\equiv A_1 + A_2,
\end{aligned}$$

where $A_1 = \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K^2\left(\frac{\tau_t - \tau}{h_j}\right) L^2\left(\frac{x_t - x}{h_j}\right) e_{t+j}^2$ and

$$A_2 = \frac{2}{nh_j^{d+1}} \sum_{t=2}^n \sum_{s=1}^{t-1} K\left(\frac{\tau_t - \tau}{h_j}\right) K\left(\frac{\tau_s - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right) L\left(\frac{x_s - x}{h_j}\right) e_{t+j} e_{s+j}.$$

By iterated expectations and change of variables, we can show that

$$\begin{aligned}
 \mathbb{E}[A_1] &= \frac{1}{nh_j^{d+1}} \sum_{t=1}^n K^2 \left(\frac{\tau_t - \tau}{h_j} \right) \mathbb{E} \left[L^2 \left(\frac{x_t - x}{h_j} \right) e_{t+j}^2 \right] \\
 &= \frac{1}{h_j^{d+1}} K^2 \left(\frac{\tau_t - \tau}{h_j} \right) \mathbb{E} \left[L^2 \left(\frac{x_t - x}{h_j} \right) \mathbb{E}[e_{t+j}^2 | x_t = x] \right] \\
 &= \frac{\sigma_j^2(x)}{h_j^{d+1}} K^2 \left(\frac{\tau_t - \tau}{h_j} \right) \mathbb{E} \left[L^2 \left(\frac{x_t - x}{h_j} \right) \right] \\
 &= \frac{\sigma_j^2(x)}{h_j^{d+1}} K^2 \left(\frac{\tau_t - \tau}{h_j} \right) \mathbb{E} \left[L^2 \left(\frac{x_t(\tau_t) - x}{h_j} \right) \right] (1 + o_P(1)) \\
 &= \frac{\sigma_j^2(x)}{h_j^{d+1}} K^2 \left(\frac{\tau_t - \tau}{h_j} \right) \int L^2 \left(\frac{x_y - x}{h_j} \right) f(\tau_t, x_y) dx_y (1 + o_P(1)) \\
 &= \frac{\sigma_j^2(x)}{h_j^{d+1}} \iint K^2 \left(\frac{\tau_y - \tau}{h_j} \right) L^2 \left(\frac{x_y - x}{h_j} \right) f(\tau_y, x_y) d\tau_y dx_y (1 + o_P(1)) \\
 &= \sigma_j^2(x) \int \cdots \int K^2(p) \prod_{i=1}^d K^2(q_i) f(\tau, x) dp dq^1 \cdots dq^d \\
 &\quad + \sigma_j^2(x) \int \cdots \int K^2(p) \prod_{i=1}^d K^2(q_i) \left(ph_j \frac{\partial f(\tau, x)}{\partial \tau} + \sum_{i=1}^d q_i h_j \frac{\partial f(\tau, x)}{\partial x^i} \right) dp dq^1 \cdots dq^d \\
 &\quad + \sigma_j^2(x) \int \cdots \int K^2(p) \prod_{i=1}^d K^2(q_i) \frac{1}{2} \left(p^2 h_j^2 \frac{\partial^2 f(\tau, x)}{\partial \tau^2} + \sum_{i=1}^d q_i^2 h_j^2 \frac{\partial^2 f(\tau, x)}{\partial x^{i^2}} \right) dp dq^1 \cdots dq^d \\
 &\quad + \sigma_j^2(x) \int \cdots \int K^2(p) \prod_{i=1}^d K^2(q_i) \frac{1}{2} \left(\sum_{i=1}^d p q_i h_j^2 \frac{\partial^2 f(\tau, x)}{\partial \tau \partial x^i} \right. \\
 &\quad \left. + 2 \sum_{i=2}^d \sum_{s=1}^{i-1} q_i q_s h_j^2 \frac{\partial^2 f(\tau, x)}{\partial x^i \partial x^s} \right) dp dq^1 \cdots dq^d + o_P(h_j^2) = \sigma_j^2(x) f(\tau, x) \kappa_0^{d+1} + O_P(h_j). \quad (\text{A15})
 \end{aligned}$$

Meanwhile, by the same steps as in Theorem 1 of Hansen (2008), we have that $\mathbb{E}[A_2] = o_P(1)$. Therefore, we can obtain that $\text{Var}[V_j(\tau, x)] = \sigma_j^2(x) f(\tau, x) \kappa_0^{d+1} + o_P(1)$.

We then use the small-block and big-block arguments (refer to Fan and Yao, 2003), that is, decompose $V_j(\tau, x)$ alternately into big blocks and small blocks, we can neglect the small blocks and use the mixing conditions to replace the big blocks by independent random variables. Then apply a Lindeberg theorem, we can get that $V_j(\tau, x) \rightarrow_D N(0, \kappa_0^{d+1} \sigma_j^2(x) f(\tau, x))$. The proof is in the same spirit as that for the standard strictly stationary setting.

By similar argument as the (A12), we have that

$$\widehat{\mathbb{E}}\widehat{f}(\tau, x) = \widehat{\mathbb{E}}\widehat{f}^*(\tau, x)(1 + o_P(1)).$$

Then the bias of $\widehat{f}(\tau, x)$ will be

$$\widehat{\mathbb{E}}\widehat{f}(\tau, x) - f(\tau, x) = \frac{\kappa_2}{2} h_j^2 \left(\frac{\partial^2 f(\tau, x)}{\partial \tau^2} + \sum_{i=1}^d \frac{\partial^2 f(\tau, x)}{\partial x^{i^2}} \right) + o_P(h_j^2). \quad (\text{A16})$$

Following the similar steps of the proof of Theorem 1.1 in Li and Racine (2007), we can obtain the variance of $\widehat{f}(\tau, x)$:

$$\text{Var} \left(\widehat{f}(\tau, x) \right) = \frac{1}{nh_j^{d+1}} \left(\kappa_0^{d+1} f(\tau, x) + O_P(h_j) \right). \quad (\text{A17})$$

Based on Eqs. (A16) and (A17) and Assumption A.1.4(ii), we can obtain that $\widehat{f}(\tau, x) - f(\tau, x) = o_P(1)$. It is also straightforward to see that $\widehat{f}(\tau, x)^{-1} = O_P(1)$.

Then $V_j(\tau, x)/\widehat{f}(\tau, x) \rightarrow_D N(0, V_{j,\tau,x})$, where $V_{j,\tau,x} = \kappa_0^{d+1} \sigma_j^2(x)/f(\tau, x)$. Combining with Eq. (A14), we have

$$\sqrt{nh_j^{d+1}} (\widehat{g}_j(\tau, x) - g_j(\tau, x) - B_j(\tau, x)) \rightarrow_D N(0, V_{j,\tau,x}), \quad (\text{A18})$$

Therefore, we have completed the proof of Theorem 2.1. $\square$

Proof of Theorem 2.2. Observe that

$$\begin{aligned} (\widehat{g}_j(\tau, x) - g_j(\tau, x)) &= \left(nh_j^{d+1}\right)^{-1/2} \widehat{f}(\tau, x)^{-1} (V_j(\tau, x) + B_j(\tau, x)) \\ &= \left(nh_j^{d+1}\right)^{-1/2} f(\tau, x)^{-1} (1 + o_P(1)) \rho_j^{-(d+1)/2} V_j(\tau, x) \\ &\quad + \left(nh_j^{d+1}\right)^{-1/2} f(\tau, x)^{-1} (1 + o_P(1)) \rho_j^{-(d+1)/2} B_j(\tau, x), \end{aligned} \quad (\text{A19})$$

which gives

$$\begin{aligned} &\left(\sum_{j=1}^J \widehat{g}_j(\tau, x) - \sum_{j=1}^J g_j(\tau, x) \right) \\ &= \left(nh^{d+1}\right)^{-1/2} f(\tau, x)^{-1} (1 + o_P(1)) \sum_{j=1}^J \rho_j^{-(d+1)/2} V_j(\tau, x) \\ &\quad + \left(nh^{d+1}\right)^{-1/2} f(\tau, x)^{-1} (1 + o_P(1)) \sum_{j=1}^J \rho_j^{-(d+1)/2} B_j(\tau, x) \\ &\equiv \left(nh^{d+1}\right)^{-1/2} f(\tau, x)^{-1} (1 + o_P(1)) S_{nJ}(\tau, x) + \left(nh^{d+1}\right)^{-1/2} f(\tau, x)^{-1} (1 + o_P(1)) R_{nJ}(\tau, x), \end{aligned} \quad (\text{A20})$$

where

$$\begin{aligned} S_{nJ}(\tau, x) &= \sum_{j=1}^J \rho_j^{-(d+1)/2} V_j(\tau, x) \\ &= \left(nh^{d+1}\right)^{-1/2} \sum_{t=1}^n \left(\sum_{j=1}^J \rho_j^{-(d+1)} K\left(\frac{\tau_t - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right) e_{t+j} \right), \end{aligned} \quad (\text{A21})$$

$$\begin{aligned} R_{nJ}(\tau, x) &= \sum_{j=1}^J \rho_j^{-(d+1)/2} B_j(\tau, x) \\ &= \left(nh^{d+1}\right)^{-1/2} \sum_{t=1}^n \left(\sum_{j=1}^J \rho_j^{-(d+1)} K\left(\frac{\tau_t - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right) (g_j(\tau_t, x_t) - g_j(\tau, x)) \right). \end{aligned} \quad (\text{A22})$$

It is obvious that $\mathbb{E}[S_{nJ}(\tau, x)] = 0$. It can be also shown that

$$\begin{aligned} S_{nJ}(\tau, x) &= \sum_{j=1}^J \rho_j^{-\frac{d+1}{2}} V_j(\tau, x) \\ &= (nh^{d+1})^{-\frac{1}{2}} \sum_{j=1}^J \rho_j^{-(d+1)} \sum_{t=1}^n K\left(\frac{\tau_t - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right) e_{t+j} \\ &= (nh^{d+1})^{-\frac{1}{2}} T_{nJ}(\tau, x), \end{aligned}$$

where $T_{nj}(\tau, x) = \sum_{t=1}^n \left(\sum_{j=1}^J \rho_j^{-(d+1)} K\left(\frac{\tau_t - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right) e_{t+j} \right) = \sum_{t=1}^n U_t(J)$, in which

$$U_t(J) = \sum_{j=1}^J \rho_j^{-(d+1)} K\left(\frac{\tau_t - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_j}\right) e_{t+j}.$$

It is easy to see that

$$T_{nj}^2(\tau, x) = \sum_{t=1}^n U_t^2(J) + 2 \sum_{t=2}^n \sum_{s=1}^{t-1} U_t(J) U_s(J),$$

and

$$\begin{aligned} U_t^2(J) &= \sum_{j=1}^J \rho_j^{-2(d+1)} K^2\left(\frac{\tau_t - \tau}{h_j}\right) L^2\left(\frac{x_t - x}{h_j}\right) e_{t+j}^2 \\ &\quad + 2 \sum_{i=2}^J \sum_{j=1}^{i-1} \rho_i^{-(d+1)} \rho_j^{-(d+1)} K\left(\frac{\tau_t - \tau}{h_i}\right) K\left(\frac{\tau_t - \tau}{h_j}\right) L\left(\frac{x_t - x}{h_i}\right) L\left(\frac{x_t - x}{h_j}\right) e_{t+i} e_{t+j}. \end{aligned}$$

According to Eq. (A15), we can show that

$$\begin{aligned} \mathbb{E}[U_t^2(J)] &= \sum_{j=1}^J \rho_j^{-2(d+1)} K^2\left(\frac{\tau_t - \tau}{h_j}\right) \mathbb{E}\left[\mathbb{E}[e_{t+j}^2 | x_t = x] L^2\left(\frac{x_t - x}{h_j}\right)\right] \\ &= \sum_{j=1}^J \rho_j^{-2(d+1)} \sigma_j^2(x) h_j^{d+1} f(\tau, x) \kappa_0^{d+1} (1 + o_P(1)) \\ &= f(\tau, x) \kappa_0^{d+1} h^{d+1} \sum_{j=1}^J \sigma_j^2(x) \rho_j^{-(d+1)} (1 + o_P(1)). \end{aligned}$$

Similar to the derivation of variance of $V_j(\tau, x)$ in Theorem 2.1, we have

$$\begin{aligned} \text{Var}(T_{nj}^2(\tau, x)) &= \mathbb{E}[T_{nj}^2(\tau, x)] = \mathbb{E}\left[\sum_{t=1}^n U_t^2(J)\right] + o_P(1) \\ &= n f(\tau, x) \kappa_0^{d+1} h^{d+1} \sum_{j=1}^J \sigma_j^2(x) \rho_j^{-(d+1)} + o_P(1). \end{aligned}$$

Hence, we have

$$\begin{aligned} \text{Var}(S_{nj}(\tau, x)) &= \mathbb{E}[S_{nj}^2(\tau, x)] = (n h^{d+1})^{-1} \mathbb{E}[T_{nj}^2(\tau, x)] \\ &= n^{-1} \frac{1}{h^{d+1}} n f(\tau, x) \kappa_0^{d+1} h^{d+1} \sum_{j=1}^J \sigma_j^2(x) \rho_j^{-(d+1)} + o_P(1) \\ &= f(\tau, x) \kappa_0^{d+1} \sum_{j=1}^J \sigma_j^2(x) \rho_j^{-(d+1)} + o_P(1). \end{aligned}$$

In view of the α -mixing condition, using the big-blocks and small-blocks arguments (see, e.g., Appendix A of Gao (2007)), we can show that as $n \rightarrow \infty$

$$\left(\sum_{j=1}^J \rho_j^{-(d+1)} \sigma_j^2(x) \right)^{-1/2} S_{nj}(\tau, x) \rightarrow_D N\left(0, f(\tau, x) \kappa_0^{d+1}\right). \quad (\text{A23})$$

From Eq. (A20), we have that

$$f(\tau, x) \sqrt{nh^{d+1}} \left(\sum_{j=1}^J \widehat{g}_j(\tau, x) - \sum_{j=1}^J g_j(\tau, x) \right) = (1 + o_P(1)) S_{nj}(\tau, x) + (1 + o_P(1)) R_{nj}(\tau, x)$$

Let $B_J(\tau, x) = \frac{R_{nj}(\tau, x)}{f(\tau, x) \sqrt{nh^{d+1}}}$. Then based on Eq. (A14) and under Assumption A.1.4(ii), it is easy to show that

$$\begin{aligned} B_J(\tau, x) &= \sum_{j=1}^J (h_j^2 R_j(\tau, x) + h_j^2 b_j(\tau, x)) \\ &= \sum_{j=1}^J \rho_j^2 h^2 \kappa_2 \left[\frac{1}{2} \frac{\partial^2 g_j(\tau, x)}{\partial \tau^2} + \frac{1}{2} \sum_{i=1}^d \frac{\partial^2 g_j(\tau, x)}{\partial x^{i2}} + f^{-1}(\tau, x) \frac{\partial f(\tau, x)}{\partial \tau} \frac{\partial g_j(\tau, x)}{\partial \tau} \right. \\ &\quad \left. + f^{-1}(\tau, x) \sum_{i=1}^d \frac{\partial f(\tau, x)}{\partial x^i} \frac{\partial g_j(\tau, x)}{\partial x^i} \right]. \end{aligned}$$

Then we have

$$f(\tau, x) \sqrt{nh^{d+1}} \left(\sum_{j=1}^J \widehat{g}_j(\tau, x) - \sum_{j=1}^J g_j(\tau, x) - B_J(\tau, x) \right) = (1 + o_P(1)) S_{nj}(\tau, x),$$

which shows that

$$f(\tau, x) \sqrt{nh^{d+1} \Sigma_J^{-1}(x)} \left(\sum_{j=1}^J \widehat{g}_j(\tau, x) - \sum_{j=1}^J g_j(\tau, x) - B_J(\tau, x) \right) \rightarrow_D N \left(0, f(\tau, x) \kappa_0^{d+1} \right),$$

where $\Sigma_J(x) = \sum_{j=1}^J \rho_j^{-(d+1)} \sigma_j^2(x)$.

Therefore, we have

$$\sqrt{nh^{d+1} \Sigma_J^{-1}(x)} \left(\sum_{j=1}^J \widehat{g}_j(\tau, x) - \sum_{j=1}^J g_j(\tau, x) - B_J(\tau, x) \right) \rightarrow_D N(0, V(\tau, x)),$$

where $V(\tau, x) = \kappa_0^{d+1} f^{-1}(\tau, x)$.

Therefore, we have completed the proof of Theorem 2.2. □